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传统 中医药 毒性 预测 基于 异构 网络

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文章 信息

关键词：中医药
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a b s t r a c t

and identifying toxic and effective clinical usage of Traditional Chinese Medicines (TCMs) gains increasing international attention for their herb ingredients are crucial for their safe use. Traditional wet-lab based pipelines for assessing herb toxicity distinct therapeutic effects. Precisely understanding the herb (TCM) toxicity ,are complex and time-consuming. Given the large volume toxicity data of ingredients ,

building computational but often fails to effectively models for efficient ingredient-based toxicity prediction and evaluation is promising we ,predict the toxicity of herbs due to the complexity herbs and toxic mechanisms. To address these challenges , , propose a heterogeneous network based approach (HerbToxNet) for predicting herb toxicity. HerbToxNet first and leverages a constructs a heterogeneous network composed with herbs ingredients and molecular targets ,,heterogeneous graph attention network to learn the representation of herbs. Meanwhile ,

it performs contrastive learning with dynamic coefficient to refine the representation by pulling close the herbs with shared toxic labels ,while pushing away the others. Next and further introduces a weighted label fusion strategy that uses toxic labels of , it uses Multilayer Perceptron (MLP) on the herb representation to predict similar herbs to augment the predicted labels of this herb. HerbToxNet outperforms competitive methods and the toxic labels of this herb ,finds out novel potential toxicities ,

with 96% toxicity labels confirmed for canonical herbs. It can mine related , and dissect the molecular mechanism of herb toxicity with toxic ingredients and targets in an interpretable way authenticity.

1. 引言

中药（草药）在世界传统医学实践中有着悠久的历史，特别是在亚洲，它一直是数百年医疗保健的基石（胡等人，2025年）。草药含有多种生物活性成分，如生物碱、黄酮类和萜类，这些成分有助于其治疗效果。尽管它们被认为是安全自然的，但许多草药与不良反应有关，包括肝毒性、肾毒性和心脏毒性等（江等人，2010年）。这些毒性效应可能由剂量不当、长期使用、草药相互作用或重金属和农药污染引起。此外，由于种植、采收和加工方法的不同，草药质量的差异进一步增加了其安全性评估的复杂性（刘等人，2020年）。鉴于这些多方面的挑战，迫切需要开发高效可靠的系统方法来系统地识别和评估草药毒性。

和 可靠的 方法 来 系统 地 识别 和 评估 草药毒性。

与化学药物简单的成分不同，一种草药含有多种生物活性成分，具有多靶点、多通路和多效应相互作用，其毒性机制更为复杂多样（吴等人，2021）。因此，草药毒性研究面临一些独特的挑战。首先，草药毒性研究需要复杂且昂贵的实验。来自不同文献的数据往往表现出显著的异质性和不一致性，而临床数据分散在各个机构，缺乏标准化格式和共享平台。这使得获取高质量毒性数据变得困难（陈，2003）。其次，草药的多成分特性和复杂的毒性机制对模型提出了更高的要求。然而，传统机器学习和深度学习方法难以处理草药的多模态特征（王等人，2004）。第三，草药

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Traditional Chinese medicine toxicity prediction by heterogeneous network

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A R T I C L E I N F O

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Heterogeneous network
Contrastive learning

A B S T R A C T

The clinical usage of Traditional Chinese Medicines (TCMs) gains increasing international attention for their distinct therapeutic effects. Precisely understanding the herb (TCM) toxicity, and identifying toxic and effective herb ingredients are crucial for their safe use. Traditional wet-lab based pipelines for assessing herb toxicity are complex and time-consuming. Given the large volume toxicity data of ingredients, building computational models for efficient ingredient-based toxicity prediction and evaluation is promising, but often fails to effectively predict the toxicity of herbs, due to the complexity herbs and toxic mechanisms. To address these challenges, we propose a heterogeneous network based approach (HerbToxNet) for predicting herb toxicity. HerbToxNet first constructs a heterogeneous network composed with herbs, ingredients and molecular targets, and leverages a heterogeneous graph attention network to learn the representation of herbs. Meanwhile, it performs contrastive learning with dynamic coefficient to refine the representation by pulling close the herbs with shared toxic labels, while pushing away the others. Next, it uses Multilayer Perceptron (MLP) on the herb representation to predict the toxic labels of this herb, and further introduces a weighted label fusion strategy that uses toxic labels of similar herbs to augment the predicted labels of this herb. HerbToxNet outperforms competitive methods and finds out novel potential toxicities, with 96% toxicity labels confirmed for canonical herbs. It can mine related toxic ingredients and targets in an interpretable way, and dissect the molecular mechanism of herb toxicity with authenticity.

1. Introduction

Herb medicine (Herb) has a long history in traditional medical practices around the world, especially in Asia, where it has been a cornerstone of healthcare for centuries (Hu et al., 2025). Herbs consist of a variety of bioactive ingredients, such as alkaloids, flavonoids, and terpenes, which contribute to their therapeutic effects. Although they are considered as safe and natural, many herbs have been linked to adverse effects, including hepatotoxicity, nephrotoxicity, cardiotoxicity, etc. (Jiang et al., 2010). These toxic effects can be caused by improper dosing, prolonged use, herb interactions, or contamination with heavy metals and pesticides. In addition, variations in herb quality arising from differences in cultivation, harvesting, and processing methods cause further complexity into their safety evaluation (Liu et al., 2020). Given these multifaceted challenges, there is an urgent need to develop efficient and reliable methodologies to systematically identify and evaluate herb toxicity.

Different from the simple composition of chemical drugs, a herb contains a variety of bioactive ingredients with multi-target, multi-pathway and multi-effect interactions, and its toxicity mechanisms are more complex and diverse (Wu et al., 2021). Therefore, the study of herb toxicity faces some distinctive challenges. First, herb toxicity studies require complex and costly experiments. Data from different literature often exhibit substantial heterogeneity and inconsistency, while clinical data are scattered across institutions, lacking standardized formats and shared platforms. This makes it difficult to obtain high-quality toxicity data (Chan, 2003). Second, the multi-ingredient nature and complex toxicity mechanisms of herbs impose higher demands on models. However, traditional machine learning and deep learning methods struggle to process the multi-modal features of herbs (Wang et al., 2004). Third, herb

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毒性预测通常简化为二元分类任务，而它本质上应该是多标签分类任务，因为一种草药通常与不同的毒性效应相关联。在我们的收集到的中医药毒性数据集中，超过一半的草药（252种中的141种）与两个或多个毒性标签相关联，这突出了多标签学习基于草药毒性预测的复杂性和必要性。第四，深度学习模型通常被视为"黑箱"，使其难以为其毒性预测提供清晰的生物学解释（高等人，2025）。这种缺乏可解释性阻碍了研究人员对毒性机制的理解，从而削弱了深度模型的真实性和临床适用性。

监管机构和研究人员一直强调提高草药产品质量的重要性（Jitˇareanu等人，2022）。评估草药毒性的传统方法，如体内动物实验和体外细胞实验，不仅耗时且成本高昂，而且还引发伦理问题。此外，鉴于成分数量庞大及其多样的作用机制，综合毒性评估仍然是一项艰巨的任务。随着高通量技术的出现以及大规模生物和化学数据集的日益可用，计算方法在草药毒性预测中获得了关注。这些方法包括机器学习模型，如随机森林、支持向量机以及深度学习框架，它们可以分析复杂、多维数据以预测毒性结果（He等人，2019a；Sun等人，2019；Yang等人，2021）。通过整合化学结构、药理活性以及已知的毒理学特征数据，这些模型为识别草药的潜在毒性并促进其安全使用提供了成本效益高且可扩展的解决方案（Jia等人，2024；Wang等人，2020；Wu等人，2019a；Yu等人，2025）。然而，现有的计算方法仍然面临由草药毒性相关数据的异质性和不完整性引起的重大挑战。大多数当前模型仅限于单一类型的信息，如分子结构或药理活性，并且通常缺乏有效整合反映草药复杂生物机制的多源数据的能力。此外，缺乏能够捕捉草药、成分、靶点和毒性之间多级相互作用的系统框架，限制了毒性预测的可解释性和真实性。鉴于这些情况，迫切需要真实的计算方法，这些方法能够系统地融合多源信息，并解决草药毒性预测中数据异质性、稀疏性和可解释性方面的挑战。

在本文中，我们通过整理现有草药毒性研究的结果构建了一个草药多标签毒性数据集，以解决分散且稀疏的草药毒性数据问题（参见第4节）。为了解决建模复杂草药的困难以及缺乏可解释性的问题，我们提出了一种基于异构网络的草药毒性预测方法（HerbToxNet）。HerbToxNet首先利用药理性质、毒理学特征以及草药之间的关联，构建了一个草药-成分-靶点异构网络，然后使用异构图注意力网络（HAN）（王等人，2019）来学习草药的表示。同时，它引入了具有动态系数的对比学习，以拉近具有共享毒性标签的草药对之间的表示，同时推远具有不同毒性标签的草药对。接下来，它使用多层感知器（MLP）在草药表示上预测初始毒性标签，并进一步引入了一种加权标签融合策略，该策略检索相似草药并合并它们的毒性标签，以细化预测标签。O

ur con tributi onsaresummar ized as below:

(i) We 构建 并 共享 一个 多标签 毒性 数据集 ，通过 收集 草药 毒性 研究 的 结果 ，这些研究散布在 不同 的草药 数据库 和 已发表 文献中。

(ii) 我们 提出 一个 统一的 框架 HerbToxNet 来 建模 草药、 成分、靶点之间的 多级 关系 ， 通过 异构网络， 这 允许 对毒性相关 机制进行 透明 分析 ， 并 增强 可解释性。

(iii) HerbToxNet 结合 异构网络建模、 对比学习、 加权标签融合来 解决 毒性数据不足的问题， 并以 合理且 真实的方式 实现毒性预测。

(iv) 大量实验 表明 HerbToxNet 优于 竞争性 方法， 识别 了 候选草药的潜在毒性 （96% 已确认）， 并 可以直接 揭示毒性相关 成分 和<style id='31'> 靶标<style id='33'>， 从<style id='35'> 一个可解释的<style id='37'> 视角。

本文的其余部分组织如下。第2节讨论相关工作，第3节详细阐述所提出的HerbToxNet框架。第4节介绍实验设置、结果和分析。第5节总结我们的工作并概述未来的努力。

2. 相关工作

现有的草药毒性计算方法通常预测一些主要的毒性类型，如肝毒性、肾毒性、心脏毒性等，并将它们建模为二元分类任务（感兴趣的毒性是否存在）。基于草药毒性的特点，发展了一系列草药毒性预测的理论和方法，可以分为三类：基于定量构效关系（QSAR）的草药毒性预测、网络毒理学、草药属性理论。

在计算毒理学中，化合物的虚拟结构通常用于通过计算方法分析、模拟、可视化或预测化学品的毒性（Raies & Bajic, 2016）。QSAR可以定量地捕捉分子结构与生物效应之间的关系，它使化合物毒性的计算预测成为可能，并已广泛应用于化学药物毒性评估（Cherkasov等人，2014）。因此，基于QSAR的方法可以适应计算预测每种草药中（类似药物）单个成分的毒性，为估计草药的整体毒性提供依据（Gao等人，2019）。何等人（2019a）收集了肝毒性草药和成分，分析了这些成分的结构，发现生物碱和萜类化合物是主要的肝毒性成分。杨等人（2021）使用QSAR评估了决明子中成分的潜在毒性，并通过线性模型阐明了结构特征与细胞毒性的关系，发现决明子中有三种成分具有肝毒性，八种成分具有肾毒性。孙等人（2019）使用草药肾毒性数据构建了QSAR模型，并使用人工神经网络（ANN）和支持向量机算法预测肾毒性。虽然基于QSAR的方法能够在分子水平上定量分析结构-毒性关系，但它们在草药毒性预测中的准确性和适用性受限于结构数据和毒理学数据的可用性和质量，以及许多成分的不完整表征。

网络毒理学，是基于网络药理学（Li等人，2019）研究生物医药的重要分支。网络毒理学的主要研究目的是通过特定的毒性-靶点-成分-药材相互作用网络分析药材毒性。一般分析过程包括：识别药材毒性成分、预测毒性成分及毒性靶点功能分析。吴等人（2019a）结合四种机器学习方法与四种不同的指纹图谱构建了16个单一分类器以预测潜在的肝毒性成分，并通过系统药理学方法探索了药材的肝毒性机制。余等人（2025）在正清风宁引起的肝损伤数据集中比较了十种机器学习模型，以找到预测肝损伤最有效的模型。虽然网络毒理学可以系统地阐明毒性机制并识别关键靶点，但这种方法的局限性在于不能直接量化药材水平毒性，且受限于不完整的药材-成分-靶点关联，导致预测结果存在偏差或不完整。

"四性五味"（亦称"四气五味"）和经度趋向构成药材药理学理论框架的关键组成部分（乔等人，2022）。草药的"四性"分为五种类型：寒、热、

toxicity prediction is often simplified as a binary classification task, whereas it should inherently be a multi-label classification task, as a herb is often associated with different toxic effects. In our collected TCM toxicity dataset, more than half of the herbs (141 out of 252) are linked with two or more toxic labels, highlighting the complexity and necessity of multi-label learning based herb toxicity prediction. Fourth, deep learning models are often seen as "black boxes", making them difficult to provide clear biological explanations for their toxicity prediction (Gao et al., 2025). This lack of interpretability hinders researchers' understanding of toxicity mechanisms, and thus undermines the authenticity and clinical applicability of deep models.

Regulatory authorities and researchers have been emphasizing the importance of improving the safety and quality control of herb products (Jităreanu et al., 2022). Traditional methods for assessing herb toxicity, such as in vivo animal testing and in vitro cellular assays, are not only time-consuming and costly but also raise ethical concerns. Moreover, given the vast number of ingredients and their diverse action mechanisms, the comprehensive toxicity assessments remain a daunting task. With the advent of high-throughput technologies and the growing availability of large-scale biological and chemical datasets, computational approaches have gained attraction in herb toxicity prediction. These approaches include machine learning models, such as random forests, support vector machines, and deep learning frameworks, which can analyze complex, multidimensional data to predict toxicity outcomes (He et al., 2019a; Sun et al., 2019; Yang et al., 2021). By integrating data on chemical structures, pharmacological activities, and known toxicological profiles, these models offer a cost-effective and scalable solution for identifying potential toxicity of herbs and facilitating their safe usage (Jia et al., 2024; Wang et al., 2020; Wu et al., 2019a; Yu et al., 2025). However, existing computational approaches still encounter significant challenges arising from the heterogeneity and incompleteness of herb toxicity related data. Most current models are limited to a single type of information, such as molecular structure or pharmacological activity, and often lack the capability to effectively integrate diverse data sources that reflect the complex biological mechanisms of herbs. Furthermore, the absence of systematic frameworks capable of capturing multi-level interactions among herbs, ingredients, targets, and toxicities, restricts both the interpretability and authenticity of toxicity predictions. Given these, there is a critical need of authentic computational approaches that can systematically fuse multi-source information and address the challenges of data heterogeneity, sparsity, and interpretability in herb toxicity prediction.

In this paper, we construct a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies to solve the problem of dispersed and scarce herb toxicity data (see Section 4). To address the difficulties in modeling complex herbs and the lack of interpretability, we propose a heterogeneous network based approach (HerbToxNet) for herb toxicity prediction. HerbToxNet first constructs a herb-ingredient-target heterogeneous network using the pharmacological properties, toxicological characteristics, and associations between herbs, and then uses Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) to learn the representations of herbs. At the same time, it introduces contrastive learning with dynamic coefficient to pull closer the representations of herb pairs with shared toxic labels while pushing apart those with different toxic labels. Next, it uses a Multilayer Perceptron (MLP) on the herb representation to predict the initial toxic labels, and further introduces a weighted label fusion strategy that retrieves similar herbs and merges their toxic labels to refine the predicted labels.

Our contributions are summarized as below:

(i) We construct and share a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies scattered in different herb databases and published literature.

(ii) We propose a unified framework HerbToxNet to model multi-level relationships among herbs, ingredients, and targets via a heterogeneous network, which allows for transparent analysis of toxicity-related mechanisms and enhances interpretability.

(iii) HerbToxNet synergizes heterogeneous network modeling, contrastive learning, and weighted label fusion to remedy scarce toxicity data and achieve toxicity prediction in a sensible and authentic way.

(iv) Extensive experiments show that HerbToxNet outperforms competitive methods, identifies potential toxicities of canonical herbs (96% confirmed), and can directly uncover toxicity-related ingredients and targets from an interpretable view.

The rest of the paper is organized as follows. Section 2 discusses the related work and Section 3 elaborates on the proposed HerbToxNet framework. Section 4 presents the experimental setup, results and analysis. Section 5 summarizes our work and outlines future efforts.

2. Related work

Existing computational methods for herb toxicity usually predict some main types of toxicity, such as hepatotoxicity, nephrotoxicity, cardiotoxicity, etc., and model them as binary classification tasks (whether the interested toxicity is present or not). Based on the characteristics of herb toxicity, a series of theories and methods for herb toxicity prediction were developed, which can be divided into three categories: herb toxicity prediction based on quantitative structure-activity relationship (QSAR), network toxicology, herb property theory.

In computational toxicology, the virtual structure of a compound is often used to analyze, simulate, visualize, or predict the toxicity of a chemical by computational methods (Raies & Bajic, 2016). QSAR can quantitatively capture the relationship between molecular structure and biological effects, it empowers the computational prediction of compound toxicity and has been widely applied in the toxicity assessment of chemical drugs (Cherkasov et al., 2014). Therefore, QSAR-based methods can be adapted to computationally predict the toxicity of individual ingredients (alike drugs) within each herb, providing a basis for estimating the overall toxicity of herbs (Gao et al., 2019). He et al. (2019a) collected hepatotoxic herbs and ingredients, analyzed the structure of these ingredients and found that alkaloids and terpenoids were the main hepatotoxic ingredients. Yang et al. (2021) used QSAR to evaluate the potential toxicity of ingredients in Cassia seed, and clarified the relationship between structural characteristics and cytotoxicity by linear model, and found that three ingredients in Cassia seed had hepatotoxicity and eight ingredients had nephrotoxicity. Sun et al. (2019) used herb nephrotoxicity data to construct a QSAR model, and used ANN and SVM algorithms to predict nephrotoxicity. While QSAR-based approaches enable quantitative analysis of structure-toxicity relationships at the molecular level, their accuracy and applicability in herbal toxicity prediction are limited by the availability and quality of both structural and toxicological data, as well as the incomplete characterization of many ingredients.

Network toxicology, is an important branch for studying biomedicine based on network pharmacology (Li et al., 2019). The main research purpose of network toxicology is to analyze herb toxicity through a specific toxicity-target-ingredient-herb interaction network. The general analysis process includes: identification of toxic ingredients of herb, prediction of toxic ingredients and functional analysis of toxic targets. Wu et al. (2019a) combined four machine learning methods with four different fingerprint sets to construct 16 single classifiers to predict potential hepatotoxic ingredients, and explored the mechanism of hepatotoxicity of herbs through systematic pharmacological methods. Yu et al. (2025) compared ten machine learning models in the *Zhengqing Fengtongning*-induced liver injury dataset to find the most effective model for predicting liver injury. While network toxicology can systematically elucidate toxic mechanisms and identify key targets, this kind of methods cannot directly quantify herb-level toxicity and are limited by incomplete herb-ingredient-target associations, suffering biased or incomplete predictions.

The "Four Properties and Five Flavors" (also known as the "Four Qi and Five Tastes") and meridian tropism constitute critical components of the theoretical framework of herb pharmacology (Qiao et al., 2022). The "Four Properties" of herb are categorized into five types: cold, hot,

温、凉和中性。“五味”分为七种类型：酸、苦、甘、辛、咸、淡和涩。经度趋向是指将草药分为12个经络：肺、大肠、胃、脾、心、小肠、膀胱、肾、心包、三焦、胆和肝。王等人（2020）基于传统人工神经网络建立了一个随机游走网络，可有效解释处方信息，以预测草药处方的副作用。贾等人（2024）通过将草药毒性与其电子电离质谱数据的先进分析描述符相关联，结合可解释的机器学习模型进行毒性预测。此外，一些研究表明，草药的功效可用于分析草药毒性（杨等人，2015；朱等人，2025）。虽然草药属性理论提供了基于传统知识的描述性见解，但其定性和主观的性质与化学药物使用的标准化表示不同，即使与先进的机器学习方法结合，也会导致数据模糊和预测性能有限。

为解决上述挑战，我们构建异构图以建模草药、成分和靶点之间的复杂关系。通过利用异构图注意力网络、具有动态系数的对比学习以及加权标签融合，我们的HerbToxNet能够处理不完整数据并融合多级特征，从而有效预测草药的毒性标签。

3. 所提出的 方法

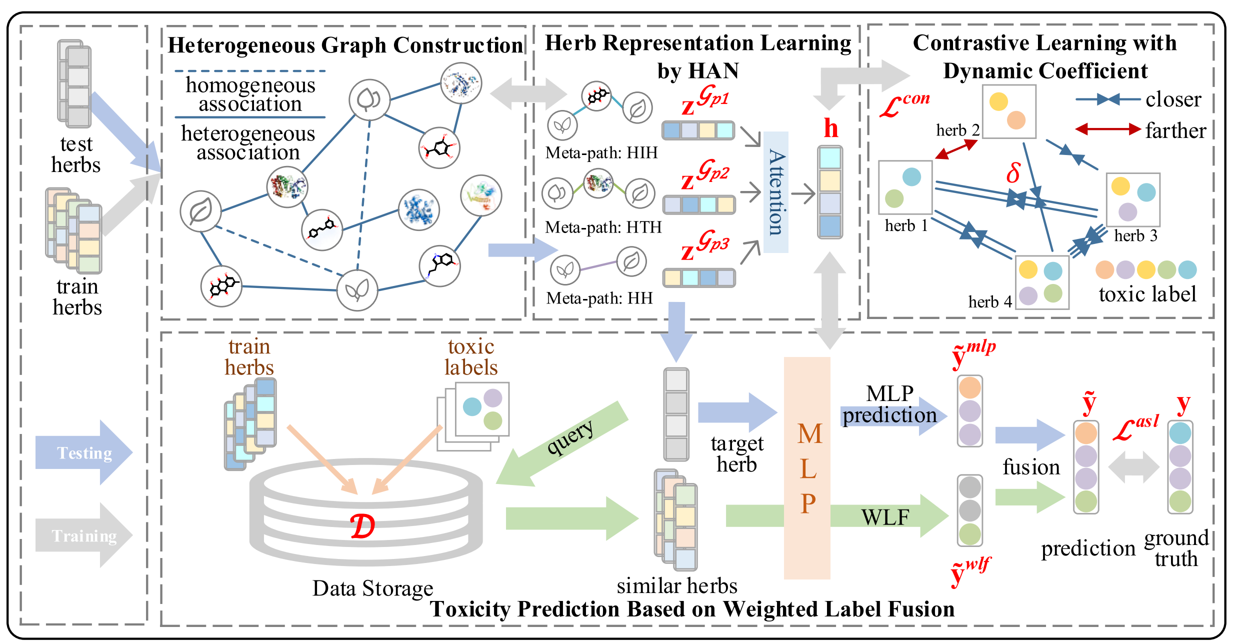
HerbToxNet 的框架如图 1 所示。HerbToxNet 包含四个主要模块：(i) 异构图构建基于草药药理性质、功效、关联性构建草药-成分-靶点异构图；(ii) 草药表示学习采用具有选定元路径的异构图注意力网络 (HAN) (王等人，2019) 提取信息丰富的草药表示；(iii) 具有动态系数的对比学习通过将具有共享毒性标签的草药聚集在一起并将具有相似标签的草药推开来优化学习到的表示；以及 (iv) 基于加权标签融合的毒性预测根据其优化表示合并 MLP 的预测和目标草药相似草药的毒性标签。这些模块按顺序组织，使 HerbToxNet 能够有效整合异构信息、学习鲁棒的表示并实现毒性预测。

毒性预测基于加权标签融合合并 MLP 的预测和目标草药相似草药的毒性标签基于其优化表示。这些模块按顺序组织，使 HerbToxNet 能够有效整合异构信息、学习鲁棒的表示并实现毒性预测。

3.1. 异构 图 构建

草药药理性质和功效数据包含对毒性预测有用的信息，但它们通常是不完整或不精确的，因为它们是基于传统文献和经验知识收集的。因此，基于草药药理性质和功效构建的现有方法往往难以利用其潜力。事实上，一种草药具有多种成分，具有协同作用，使其毒性复杂，仅凭药理性质数据难以描述和捕捉（凯撒 & 切赫，2019；弗洛拉等，2013）。类似地，网络毒理学也面临挑战，因为成分的复杂性使得现有方法难以准确预测其毒性效应。此外，这些模型通常缺乏可解释性，阻碍了对生物机制的理解，并降低了它们在实际应用中的可靠性（何等，2019a,b）。

为解决上述问题，我们将异构草药-成分-靶点图定义为$\mathcal{G} = (\mathcal{V}, \mathcal{E})$，通过联合建模成分节点和靶点节点及其相互作用，编码草药的多成分和多靶点特性。节点集$\mathcal{V} = \mathcal{V}_h \cup \mathcal{V}_c \cup \mathcal{V}_t$是草药-成分、草药-靶点、成分-靶点和草药-草药边的并集。边集 $\mathcal{E} = \mathcal{E}_{hc} \cup \mathcal{E}_{ht} \cup \mathcal{E}_{ct} \cup \mathcal{E}_{hh}$ 是草药-成分、草药-靶点、成分-靶点和草药-草药边的并集。我们旨在将每种草药的药理性质和功效整合到其节点表示中，从而将领域知识与异构图的结构信息相结合。i种草药的药理性质和功效分别表示为$\mathbf{p}_i \in \{0,1\}^{L_p}$ 和$\mathbf{e}_i \in \{0,1\}^{L_e}$，



图。 **1.** The overview of HerbToxNet framework. HerbToxNet first constructs a herb-ingredient-target heterogeneous graph and selects three types of meta-paths: ,, , , herb-ingredient-herb (HIH) herb-target-herb (HTH) and herb-herb (HH) to build the meta-path matrix. Then it utilizes the attention module of heterogeneous) , graph neural network to synthesize features ($\mathbf{z}^{\mathcal{G}_{P_1}}, \mathbf{z}^{\mathcal{G}_{P_2}}, \mathbf{z}^{\mathcal{G}_{P_3}}$ under different meta-paths to obtain the representation (**h**) of the herb. Next it introduces a contrastive) , learning loss ($\mathcal{L}^{con}$ with a dynamic coefficient δ to pull together the representations of herbs with shared toxic labels while pushing apart those without shared , , , , labels thereby optimizing herb representations. After that it predicts the toxic labels in two ways an MLP classifier to generate $\tilde{\mathbf{y}}^{mlp}$ and a weighted label fusion , strategy to aggregate labels $\tilde{\mathbf{y}}^{wlf}$ from similar herbs in the training set $\mathcal{D}$ and then merges $\tilde{\mathbf{y}}^{mlp}$ and $\tilde{\mathbf{y}}^{wlf}$ into $\tilde{\mathbf{y}}$. Finally HerbToxNet defines asymmetric classification) loss ($\mathcal{L}^{asl}$ between $\tilde{\mathbf{y}}$ and the ground-truth labels $\mathbf{y}$ to optimize the overall framework.

warm, cool, and neutral. The “Five Flavors” are classified into seven types: sour, bitter, sweet, pungent, salty, bland, and astringent. Meridian tropism refers to classifying herbs into 12 channels: lung, large intestine, stomach, spleen, heart, small intestine, bladder, kidney, pericardium, triple burner, gallbladder, and liver. Wang et al. (2020) established a random walk network that can effectively interpret prescription information on the basis of traditional ANN to predict side effects of herb prescriptions. Jia et al. (2024) made toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. In addition, some studies have shown that the efficacy of herbs can be used to analyze herb toxicity (Yang et al., 2015; Zhu et al., 2025). While herb property theory offers descriptive insights rooted in traditional knowledge, its qualitative and subjective nature, distinct from the standardized representations used for chemical drugs, results in ambiguous data and limited predictive performance even when combined with advanced machine learning methods.

To address the above challenges, we construct a heterogeneous graph to model the complex relationships between herbs, ingredient and targets. By leveraging Heterogeneous Graph Attention Networks, contrastive learning with dynamic coefficient, and weighted label fusion, our HerbToxNet can handle incomplete data and fuse multi-level features, thereby effectively predicting the toxic labels of herbs.

3. The proposed methodology

The framework of HerbToxNet is illustrated in Fig. 1. HerbToxNet comprises four main modules: (i) Heterogeneous graph construction builds a herb-ingredient-target heterogeneous graph based on herb pharmacological properties, efficacy, associations; (ii) Herb representation learning employs a Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) with selected meta-paths to extract informative herb representations; (iii) Contrastive learning with dynamic coefficient optimizes the learned representations by pulling together herbs with shared toxic labels and pushing apart those with dissimilar labels; and (iv) Toxicity prediction based on weighted label fusion merges the predictions from an MLP and toxic labels from similar herbs of the target herb based on their optimized representations. These modules are organized sequentially, enabling HerbToxNet to effectively integrate heterogeneous information, learn robust representations, and achieve toxicity prediction.

3.1. Heterogeneous graph construction

Herb pharmacological property and efficacy data contain useful information for toxicity prediction, but they are often incomplete or imprecise, as they are collected based on traditional literature and empirical knowledge. As such, existing methods built on herb pharmacological property and efficacy often struggle to exploit their potential. In fact, a herb has multiple ingredients with synergistic effects, making its toxicity complex and difficult to describe and capture with pharmacological property data alone (Caesar & Cech, 2019; Flora et al., 2013). Similarly, network toxicology faces challenges, as the complexity of ingredients makes it difficult for existing methods to accurately predict their toxic effects. Furthermore, these models often lack interpretability, hindering the understanding of biological mechanisms and reducing their reliability in practical applications (He et al., 2019a,b).

To address the above issues, we define the heterogeneous herb-ingredient-target graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, which encode the multi-ingredient and multi-target nature of herbs by jointly modeling ingredient and target nodes, as well as their interactions. The node set $\mathcal{V} = \mathcal{V}_h \cup \mathcal{V}_c \cup \mathcal{V}_t$ is the union of herb, ingredient, and target nodes, and the edge set $\mathcal{E} = \mathcal{E}_{hc} \cup \mathcal{E}_{ht} \cup \mathcal{E}_{ct} \cup \mathcal{E}_{hh}$ is the union of herb-ingredient, herb-target, ingredient-target, and herb-herb edges. We aim to integrate both the pharmacological properties and efficacy of each herb into its node representation, thereby combining domain knowledge with structural information of the heterogeneous graph. The pharmacological properties and efficacy of ith herb are denoted as $\mathbf{p}_i \in \{0,1\}^{L_p}$ and $\mathbf{e}_i \in \{0,1\}^{L_e}$,

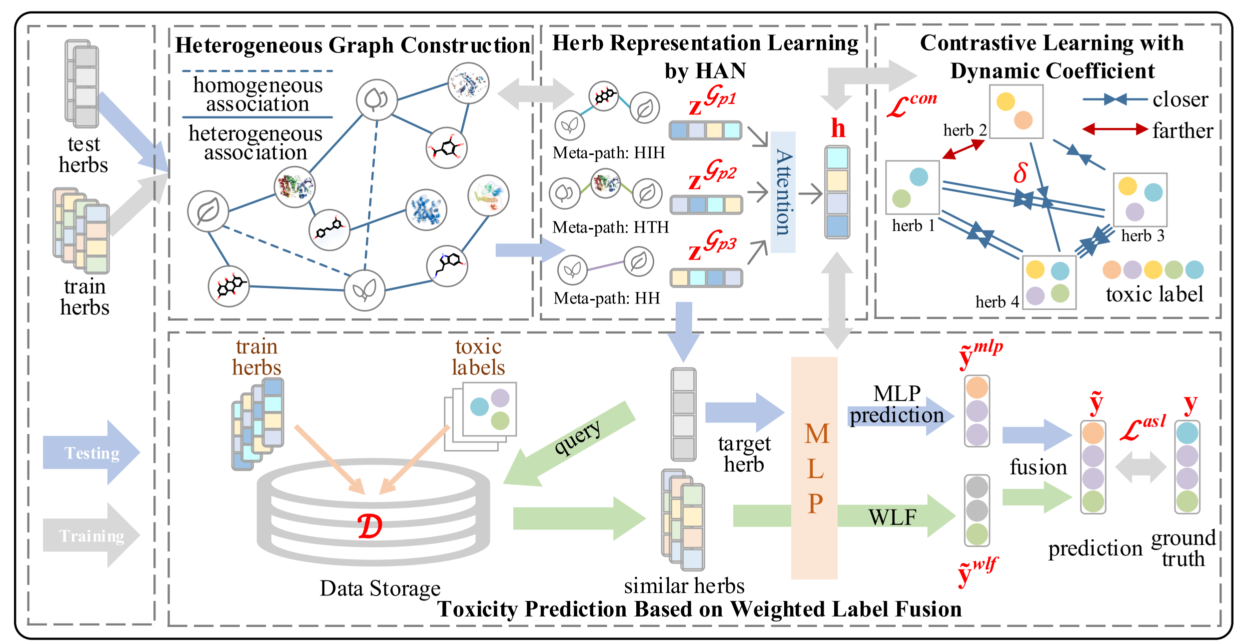


Fig. 1. The overview of HerbToxNet framework. HerbToxNet first constructs a herb-ingredient-target heterogeneous graph, and selects three types of meta-paths: herb-ingredient-herb (HIH), herb-target-herb (HTH), and herb-herb (HH) to build the meta-path matrix. Then, it utilizes the attention module of heterogeneous graph neural network to synthesize features ($\mathbf{z}^{\mathcal{G}_{P_1}}, \mathbf{z}^{\mathcal{G}_{P_2}}, \mathbf{z}^{\mathcal{G}_{P_3}}$) under different meta-paths to obtain the representation (**h**) of the herb. Next, it introduces a contrastive learning loss ($\mathcal{L}^{con}$) with a dynamic coefficient δ to pull together the representations of herbs with shared toxic labels, while pushing apart those without shared labels, thereby optimizing herb representations. After that, it predicts the toxic labels in two ways, an MLP classifier to generate $\tilde{\mathbf{y}}^{mlp}$, and a weighted label fusion strategy to aggregate labels $\tilde{\mathbf{y}}^{wlf}$ from similar herbs in the training set $\mathcal{D}$, and then merges $\tilde{\mathbf{y}}^{mlp}$ and $\tilde{\mathbf{y}}^{wlf}$ into $\tilde{\mathbf{y}}$. Finally, HerbToxNet defines asymmetric classification loss ($\mathcal{L}^{asl}$) between $\tilde{\mathbf{y}}$ and the ground-truth labels $\mathbf{y}$ to optimize the overall framework.

其中，L_p 和L_e 分别表示药理性质和功效的维度。$p_{ij} = 1$ 如果i种草药具有j种药理性质，否则为$p_{ij} = 0$ 。$e_{ij} = 1$ 如果i种草药含有j种功效，否则由$e_{ij} = 0$ 。我们使用草药的药理性质和功效作为其初始节点特征。对于i种草药，其特征表示$\mathbf{f}_i$ 是通过将$\mathbf{p}_i$ 和$\mathbf{e}_i$ 的串联进行变换获得的：

$$\mathbf{f}_i = \mathrm{ReLU}\left(\mathbf{W}_h\left(\mathbf{p}_i||\mathbf{e}_i\right) + \mathbf{b}_h\right) \tag{1}$$

where $\mathbf{W}_h$ and $\mathbf{b}_h$ 分别表示可训练权重矩阵和偏置向量。|| 表示连接操作，而ReLU(·) 是ReLU激活函数。由于我们的目标是草药毒性预测，并且异构图将基于预定义元路径使用HAN进行表示。因此，无需为成分和靶点节点分配特定的初始特征。

异构图$\mathcal{G}$ 由三种节点类型组成：草药、成分和靶点，它们通过三种类型的异构关系（草药-成分、草药-靶点和成分-靶点）以及一种类型同构关系（草药-草药）连接。成分或靶点之间的同构关系需要显式的相似性计算或手动管理，这会引入冗余和繁琐的工作，因此我们的工作中不使用这些关系。这种图结构为建模草药的结构和属性信息提供了一个统一的工具，促进了毒性预测。

3.2. 草药 表示 学习 by HAN

在异构图$\mathcal{G}$ 上给定一个元路径p ，其形式为 $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \cdots \xrightarrow{r_l} v_{l+1}$，表示节点类型$v_1$ 和v_{l+1} 通过复合关系$r = r_1 \circ r_2 \circ \cdots \circ r_l$ 连接，其中∘ 是该关系上的复合算子，$v_i \in \mathcal{V}$ 和$r_i \in \mathcal{E}$。换句话说，关系r 由那些 r_1、r_2、…、r_l 的l 关系组成。因此，元路径可以捕获异构图中的特定语义信息，不同的元路径代表不同的语义内容。

例如，Herb-Ingredient-Herb (HIH) 的元路径表明这两种草药含有相同的成分，这意味着这两种草药可能具有相似的功能和靶点关联。我们基于 HIH 元路径构建矩阵，通过将异构图中草药和成分的关联矩阵与其转置相乘。类似地，Herb-Target-Herb (HTH) 的元路径表明这两种草药与同一个靶点关联。Herb-Herb (HH) 的元路径表示两种草药之间的相似性。HTH 和 HH 矩阵的构建方法与 HIH 矩阵类似。

这里，我们将异构图$\mathcal{G}$ 分解为多个子图，每个子图$\mathcal{G}_p$ 由通过元路径p连接的所有邻居对组成，从而被指定为基于元路径的子图。具体来说，对于以草药节点开始或结束的元路径集合，我们将其表示为$\mathcal{P} = \{p_1, p_2, \ldots, p_M\}$，其中 $M = |\mathcal{P}|$ 是所考虑的元路径数量。

异构图注意力网络 (HAN) (王等人, 2019) 是一种在异构图上进行图表示学习的高效工具，能够通过分层注意力机制为不同的邻居和元路径分配不同的权重。在此背景下，我们利用 HAN 学习每个元路径诱导子图内的节点表示。HAN 自适应地聚合每个子图中节点的邻居信息，以整合节点拓扑和特征。具体来说，对于给定的节点及其关联的子图，HAN 首先采用节点级自注意力机制来学习其邻居节点的重要性（注意力分数）。然后，根据其注意力分数自适应地聚合邻居的特征表示。最后，使用语义级注意力机制来组合来自不同元路径的表示。

对于 第 k个 元路径 基于 子图 $\mathcal{G}_{p_k}$， 节点 对 的重要性 $e_{ij}^{\mathcal{G}_{p_k}}$ 可以 表示 如下： （ , j ） 可以 表示 如下：

$$e_{ij}^{\mathcal{G}_{p_k}} = \mathrm{LeakyReLU}\left(\mathbf{a}_{\mathcal{G}_{p_k}}^T \cdot \left[\mathbf{f}_i^{\mathcal{G}_{p_k}} \parallel \mathbf{f}_j^{\mathcal{G}_{p_k}}\right]\right) \tag{2}$$

其中 $\mathbf{a}_{\mathcal{G}_{p_k}}$ 是 第 个 元路径 的 注意力 参数<style id='16'> 。为了<style id='18'> 使不同<style id='20'> 元路径<style id='22'> 的<style id='24'> 注意力<style id='26'> 系数<style id='28'> 可比较<style id='30'> ， <style id='32'> 我们使用<style id='34'> softmax<style id='36'> 函数<style id='38'> 来<style id='40'> 归一化<style id='42'> $e_{ij}^{\mathcal{G}_{p_k}}$：

$$e_{ij}^{\mathcal{G}_{p_k}\prime} = \frac{\exp\left(e_{ij}^{\mathcal{G}_{p_k}}\right)}{\sum_{j' \in \mathcal{V}_{hi}^{\mathcal{G}_{p_k}}} \exp\left(e_{ij'}^{\mathcal{G}_{p_k}}\right)} \tag{3}$$

其中 $\mathcal{V}_{hi}^{\mathcal{G}_{p_k}}$ 表示第 k 个元路径中节点 i 的邻居节点集合。接下来，我们使用 ReLU 激活函数聚合所有元路径实例的特征向量，以基于第 k 个元路径推导出节点 i 的表示：

$$\mathbf{f}_i^{\mathcal{G}_{p_k}} = \mathrm{ReLU}\left(\sum_{j \in \mathcal{V}_{hi}^{\mathcal{G}_{p_k}}} e_{ij}^{\mathcal{G}_{p_k}\prime} \cdot \mathbf{f}_j^{\mathcal{G}_{p_k}}\right) \tag{4}$$

在本研究中，我们引入多头注意力机制以增强注意力系数学习过程的稳定性，并减轻单个注意力头的影响。具体而言，我们独立重复注意力机制K 次，并将每个注意力头学习到的向量表示进行连接。因此，节点i在k元路径下的表示$\mathbf{z}_i^{\mathcal{G}_{p_k}}$ 为：

$$\mathbf{z}_i^{\mathcal{G}_{p_k}} = \Big\|_{j=1}^{K} \mathbf{f}_{ij}^{\mathcal{G}_{p_k}} \tag{5}$$

通常，异构图中的每个节点包含多种类型的语义信息，而单个元路径下节点特征的聚合只能反映节点信息的一个方面。为了学习更全面的节点嵌入，整合由多个元路径表示的语义信息至关重要。从不同元路径聚合嵌入的一种简单方法是取平均值。然而，由于元路径在异构图中对节点i 的重要性不同，我们采用注意力机制为每个元路径模式分配权重，然后进行聚合。这确保了具有最重要语义信息的嵌入被赋予最大的重要性。更具体地说，我们学习每个元路径下子图的重要性，并获得$|\mathcal{P}|$ 组节点嵌入，其中每组表示对应元路径下所有节点的加权聚合：

$$w_{\mathcal{G}_{p_k}} = \frac{1}{|\mathcal{V}_h|} \sum_{i \in \mathcal{V}_h} \mathbf{q}^T \cdot \tanh\left(\mathbf{W}_k \cdot \mathbf{z}_i^{\mathcal{G}_{p_k}} + \mathbf{b}_k\right) \tag{6}$$

where $\mathbf{W}_k$, $\mathbf{b}_k$, and $\mathbf{q}_k$ 表示与第 k条元路径相关的权重矩阵、偏置向量和语义级注意力向量。tanh(·) 表示tanh激活函数。在为特定节点获取不同元路径的注意力分数后，我们使用softmax函数对它们进行归一化，从而能够在不同元路径之间对语义信息进行有意义的比较。

$$w'_{\mathcal{G}_{p_k}} = \frac{\exp(w_{\mathcal{G}_{p_k}})}{\sum_{j=1}^{|\mathcal{P}|} \exp(w_{\mathcal{G}_{p_j}})} \tag{7}$$

最后， 我们 整合 了 来自 不同 元路径 的 节点 嵌入 ， 基于 它们 respective 的 注意力 分数。 第 个 草药 节点 嵌入 $\mathbf{h}_i$ 如下：

$$\mathbf{h}_i = \sum_{k=1}^{|\mathcal{P}|} w'_{\mathcal{G}_{p_k}} \cdot \mathbf{z}_i^{\mathcal{G}_{p_k}} \tag{8}$$

respectively, where L_p and L_e represent the dimension of pharmacological properties and efficacy. $p_{ij} = 1$ if ith herb possesses the jth pharmacological properties, and $p_{ij} = 0$ otherwise. $e_{ij} = 1$ if ith herb contains the jth efficacy, and $e_{ij} = 0$ otherwise. We use the pharmacological properties and efficacy of herbs as their initial node features. For the ith herb, its feature representation $\mathbf{f}_i$ is obtained by applying a transformation to the concatenation of $\mathbf{p}_i$ and $\mathbf{e}_i$ as:

$$\mathbf{f}_i = \mathrm{ReLU}\left(\mathbf{W}_h\left(\mathbf{p}_i||\mathbf{e}_i\right) + \mathbf{b}_h\right) \tag{1}$$

where $\mathbf{W}_h$ and $\mathbf{b}_h$ denote the trainable weight matrix and bias vector, respectively. || represents the concatenation operation, and ReLU(·) is the ReLU activation function. Since our objective is herb toxicity prediction, and the heterogeneous graph will be represented using HAN based on predefined meta-paths. Therefore, there is no need to assign specific initial features to the ingredient and target nodes.

The heterogeneous graph $\mathcal{G}$ consists of three node types: herb, ingredient, and target, which are connected by three types of heterogeneous relations (herb-ingredient, herb-target, and ingredient-target), as well as one type of homogeneous relation (herb-herb). Homogeneous relations between ingredients or targets require explicit similarity computations or manual curation, which introduce redundancy and tedious efforts, so these relations are not used in our work. This graph structure provides a unified tool for modeling both the structure and attribute information of herbs, facilitating toxicity prediction.

3.2. Herb representation learning by HAN

Give a meta-path p on the heterogeneous graph $\mathcal{G}$, with the meta-path taking the form of $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \cdots \xrightarrow{r_l} v_{l+1}$, denoting that the node types v_1 and v_{l+1} are connected by a composite relationship $r = r_1 \circ r_2 \circ \cdots \circ r_l$, where ∘ is the composite operator on the relationship, $v_i \in \mathcal{V}$ and $r_i \in \mathcal{E}$. In other words, the relationship r is composed of l relationships of those $r_1, r_2, \ldots, r_l$. Thus, meta-path can capture specific semantic information within a heterogeneous graph, with different meta-paths representing different semantic contents.

For example, the meta-path of Herb-Ingredient-Herb (HIH) suggests that both herbs contain the same ingredient, implying that these two herbs may possess similar functionalities and target associations. We construct the matrix based on the HIH meta-path by multiplying the association matrix of herbs and ingredients in the heterogeneous graph with its transpose. Similarly, meta-path Herb-Target-Herb (HTH) suggests that both herbs link with the same target. The meta-path Herb-Herb (HH) indicates the similarity between two herbs. The construction methods for the HTH and HH matrices are similar to that of the HIH matrix.

Here, we decompose the heterogeneous graph $\mathcal{G}$ into multiple subgraphs, each subgraph $\mathcal{G}_p$ is composed of all neighbor pairs connected by the meta-path p, thereby designated as meta-path based subgraphs. Specifically, for the set of meta-paths that start or end with herb nodes, we denote it as $\mathcal{P} = \{p_1, p_2, \ldots, p_M\}$, where $M = |\mathcal{P}|$ is the number of meta-paths considered.

The Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) is an effective tool for graph representation learning on heterogeneous graphs, capable of assigning different weights to different neighbors and meta-paths via a hierarchical attention mechanism. In this context, we utilize HAN to learn node representations within each meta-path induced subgraph. HAN adaptively aggregates the neighbor information of nodes in each subgraph to integrate both node topology and features. Specifically, for a given node and its associated subgraph, HAN first employs a node-level self-attention mechanism to learn the importance (attention scores) of its neighboring nodes. Then, it adaptively aggregates the feature representations of neighbors based on their attention scores. Finally, a semantic-level attention mechanism is used to combine representations from different meta-paths.

For the kth meta-path based subgraph $\mathcal{G}_{p_k}$, the importance $e_{ij}^{\mathcal{G}_{p_k}}$ of node pair (i, j) can be formulated as follows:

$$e_{ij}^{\mathcal{G}_{p_k}} = \mathrm{LeakyReLU}\left(\mathbf{a}_{\mathcal{G}_{p_k}}^T \cdot \left[\mathbf{f}_i^{\mathcal{G}_{p_k}} \parallel \mathbf{f}_j^{\mathcal{G}_{p_k}}\right]\right) \tag{2}$$

where $\mathbf{a}_{\mathcal{G}_{p_k}}$ is the attention parameter of the kth meta-path. To make the attention coefficients of different meta-paths comparable, we use the softmax function to normalize $e_{ij}^{\mathcal{G}_{p_k}}$:

$$e_{ij}^{\mathcal{G}_{p_k}\prime} = \frac{\exp\left(e_{ij}^{\mathcal{G}_{p_k}}\right)}{\sum_{j' \in \mathcal{V}_{hi}^{\mathcal{G}_{p_k}}} \exp\left(e_{ij'}^{\mathcal{G}_{p_k}}\right)} \tag{3}$$

where $\mathcal{V}_{hi}^{\mathcal{G}_{p_k}}$ represents the set of neighbor nodes of node i in the kth meta-path. Next, we aggregate the feature vectors of all meta-path instances using the ReLU activation function to derive the representation of node i based on the kth meta-path:

$$\mathbf{f}_i^{\mathcal{G}_{p_k}} = \mathrm{ReLU}\left(\sum_{j \in \mathcal{V}_{hi}^{\mathcal{G}_{p_k}}} e_{ij}^{\mathcal{G}_{p_k}\prime} \cdot \mathbf{f}_j^{\mathcal{G}_{p_k}}\right) \tag{4}$$

In this study, we introduce a multi-head attention mechanism to enhance the stability of the attention coefficient learning process and mitigate the impact of individual attention heads. Specifically, we independently repeat the attention mechanism K times and concatenate the vector representations learned by each attention head. Thus, the representation $\mathbf{z}_i^{\mathcal{G}_{p_k}}$ of node i under the kth meta-path is given by:

$$\mathbf{z}_i^{\mathcal{G}_{p_k}} = \Big\|_{j=1}^{K} \mathbf{f}_{ij}^{\mathcal{G}_{p_k}} \tag{5}$$

Generally, each node in the heterogeneous graph contains multiple types of semantic information, and the aggregation of nodes features under a single meta-path can only reflect one aspect of the node's information. To learn more comprehensive node embeddings, it is essential to integrate the semantic information represented by multiple meta-paths. A simple method for aggregating embeddings from different meta-paths is to take the average. However, due to the varying importance of meta-paths to node i in the heterogeneous graph, we adopt an attention mechanism to assign weights to each meta-path pattern and then perform aggregation. This ensures that the embeddings with the most crucial semantic information are given the most significance. More specifically, we learn the importance of each meta-path based subgraph and obtain $|\mathcal{P}|$ sets of node embeddings, where each set represents the weighted aggregation of all nodes under the corresponding meta-path:

$$w_{\mathcal{G}_{p_k}} = \frac{1}{|\mathcal{V}_h|} \sum_{i \in \mathcal{V}_h} \mathbf{q}^T \cdot \tanh\left(\mathbf{W}_k \cdot \mathbf{z}_i^{\mathcal{G}_{p_k}} + \mathbf{b}_k\right) \tag{6}$$

where $\mathbf{W}_k$, $\mathbf{b}_k$, and $\mathbf{q}_k$ denote the weight matrix, bias vector, and semantic-level attention vector associated with the kth meta-path. tanh(·) represents the tanh activation function. After obtaining the attention scores of different meta-paths for a specific node, we normalize them using the softmax function, allowing for meaningful comparisons of the semantic information across different meta-paths.

$$w'_{\mathcal{G}_{p_k}} = \frac{\exp(w_{\mathcal{G}_{p_k}})}{\sum_{j=1}^{|\mathcal{P}|} \exp(w_{\mathcal{G}_{p_j}})} \tag{7}$$

Finally, we integrate the node embeddings from different meta-paths based on their respective attention scores. The ith herb node embedding $\mathbf{h}_i$ is as follows:

$$\mathbf{h}_i = \sum_{k=1}^{|\mathcal{P}|} w'_{\mathcal{G}_{p_k}} \cdot \mathbf{z}_i^{\mathcal{G}_{p_k}} \tag{8}$$

3.3. 对比 学习 与 动态 系数

上述草药表示可能无法有效捕获草药之间的毒性相关相似性和差异性，特别是在每个草药可以拥有多个毒性标签的设置下。因此，我们提出一种具有动态系数的对比学习策略来更新草药表示。基于规范对比学习的方法旨在最小化同类实例之间的距离，同时最大化不同类实例之间的距离（陈等人，2020；哈德塞尔等人，2006；科斯拉等人，2020）。然而，在多标签草药毒性预测中，两种草药通常只共享部分毒性标签，并且每种草药都拥有独特的毒性标签。这种多标签现象比单标签设置更复杂，使得简单地处理两种草药为“正”或“负”对变得困难。

为捕捉具有多毒性标签的草药之间的复杂关系，我们基于毒性标签语义相似度定义了一个动态系数。与单标签设置中的二值指示器不同，该系数提供了一种灵活的方式来数值量化草药对之间重叠标签的程度，使模型能够考虑多标签毒性预测任务中固有的复杂关系。具体来说，我们测量两个草药之间的特征相似度如下：

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{||\mathbf{h}_i||_2\, ||\mathbf{h}_j||_2} \tag{9}$$

其中 $\cos(\ldots,\ldots)$ 表示 余弦 相似度, $\mathbf{h}_i$ 和 $\mathbf{h}_j$ 分别是 第 i个和 第 j个草药 的 学习 嵌入 ， 分别 ， 和 $||\cdot||_2$ 表示 第 L_2 范数。

为 进一步 整合 毒性 标签 关系， 我们 定义 第 标签 语义 相似度 β_{ij} 为 两个草药之间共享 毒性 标签 的比例：

$$\beta_{ij} = \frac{\mathbf{y}_i^T \cdot \mathbf{y}_j}{\mathbf{y}_i^T \cdot \mathbf{y}_i} \tag{10}$$

其中 $\mathbf{y}_i \in \{0,1\}^L$ 和 $\mathbf{y}_j \in \{0,1\}^L$ 分别是第i个和第j个草药的<style id='40'> </style>多热毒性标签向量，<style id='40'> </style>而L 是毒性标签的数量。分子$\mathbf{y}_i^T \cdot \mathbf{y}_j$计算两个草药之间共享的毒性标签的数量，而分母$\mathbf{y}_i^T \cdot \mathbf{y}_i$表示第$i$个草药的毒性标签的总数。这种基于比例的语义相似度使模型不仅能够区分单标签设置中完全相同或完全不同的标签集，还能够量化多标签场景中特有的部分相似度。

在后续实验中，我们发现β 可能不适合作为对比学习的动态系数，因为它不能提供两个草药之间相关性的可靠度量。特别是，β 会为具有低毒性标签相似度的草药样本分配过大的系数。这个问题会降低草药毒性预测的性能。因此，我们引入一个可调的超参数t 来调整β ，以获得更合适的动态系数 δ:

$$\delta_{ij} = (\beta_{ij})^t \tag{11}$$

对比学习损失与动态系数δ_{ij} 对于草药对的定义如下：

$$\mathcal{L}_{ij}^{con} = -\delta_{ij} \log \frac{\exp\left(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau\right)}{\sum_{k=1,k\neq i}^{N} \exp\left(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau\right)} \tag{12}$$

其中 τ 是用于缩放相似度分数的温度超参数，而 N 是批次中草药的总数量。这种公式允许模型强调具有较高毒性相似度的相似草药对，同时有效地学习具有多个相关毒性标签的草药的表示。这种公式明确利用了草药的多标签毒性，因为每对草药对损失的贡献会根据它们共享毒性标签的程度进行缩放。

对于给定的草药对 $(\mathbf{h}_i, \mathbf{h}_j)$，更高的语义相似度$\beta_{ij}$ 会导致更大的动态系数 δ_{ij}，进而导致更大的损失 $\mathcal{L}_{ij}^{con}$。因此，特征相似度 $\cos(\mathbf{h}_i, \mathbf{h}_j)$ 会被优化到更高的值。相反，如果两个样本没有共享任何标签 ($\beta_{ij} = \delta_{ij} = 0$)，则值 $\mathcal{L}_{ij}^{con}$ 会变为零，它们的余弦相似度 $\cos(\mathbf{h}_i, \mathbf{h}_j)$ 会被优化到更低的值。对于 N 草药的对比学习损失（带动态系数）计算如下：

$$\mathcal{L}^{con} = \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \mathcal{L}_{ij}^{con} \tag{13}$$

3.4. 草药 毒性 预测 基于 加权 标签 融合

For the ith 草药, 其 表示 $\mathbf{h}_i$ 通过 一个 三层 MLP, 然后 通过 一个 sigmoid 激活 函数, 以 计算 所有 毒性 标签的概率 向量 :

$$\tilde{\mathbf{y}}_i^{mlp} = \text{sigmoid}(\text{MLP}(\mathbf{h}_i)) \tag{14}$$

where MLP(·) denotes a 三层 MLP. $\tilde{\mathbf{y}}_i^{mlp} \in \{0,1\}^L$ 表示 预测 的概率向量 ， 该向量表明 第 i 个 草药 与 毒性标签相关联。

然而，上述多层感知器可能无法充分利用训练数据中编码的宝贵知识。为了进一步增强毒性预测，我们引入了一种加权标签融合（WLF）策略。对于每种目标草药，我们检索具有相似表示的训练草药。通过对比学习，这些邻居更有可能与目标草药共享有意义的毒性模式。HerbToxNet以加权方式聚合邻近草药的毒性标签，这补充了多层感知器的输出，并有助于更准确的毒性预测。

令$\mathcal{D} = \{(\mathbf{h}_j, \mathbf{y}_j)\}_{j=1}^N$ 表示包含N 种草药的训练集。给定第i种具有表示$\mathbf{h}_i$的目标草药，我们通过计算$\mathbf{h}_i$ 与$\mathcal{D}$中的所有$\mathbf{h}_j$ 之间的余弦相似度，从训练集中检索其相关邻居。相似度超过预定义阈值σ 的邻居被选中形成邻居集：

$$\mathcal{N}_\sigma = \{(\mathbf{h}_j, \mathbf{y}_j) \mid \cos(\mathbf{h}_i, \mathbf{h}_j) \geq \sigma\} \tag{15}$$

与其使用K近邻，我们的基于阈值的策略自适应地选择与目标草药在表示空间中接近的邻居。

基于WLF的 预测 对于 i种 草药 是通过 聚合 所选邻居的 标签 向量 获得的 ， 这些向量根据 它们与 目标草药 的相似性 进行加权 ， 具体如下:

$$\tilde{\mathbf{y}}_i^{wlf} = \sum_{(\mathbf{h}_j, \mathbf{y}_j) \in \mathcal{N}_\sigma} \mu_j \mathbf{y}_j \tag{16}$$

其中 权重 μ_j 定义为:

$$\mu_j = \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau')}{\sum_{(\mathbf{h}_k, \mathbf{y}_k) \in \mathcal{N}_\sigma} \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau')} \tag{17}$$

其中τ' 是一个温度超参数，用于控制权重分布的尖锐程度。通过这种方式，与第 i 种草药相似度更高的邻居会被赋予更大的权重，这些权重对基于WLF的预测比其他邻居有更显著的影响。

为此 ， 我们 结合 基于MLP的预测 $\tilde{\mathbf{y}}_i^{mlp}$ 与 基于WLF的预测 $\tilde{\mathbf{y}}_i^{wlf}$ ， 具体如下:

$$\tilde{\mathbf{y}}_i = \alpha \tilde{\mathbf{y}}_i^{wlf} + (1-\alpha) \tilde{\mathbf{y}}_i^{mlp} \tag{18}$$

其中 标量 参数 α 平衡 了 两种预测来源。

我们采用非对称损失 (ASL) (Ridnik 等人，2021) 作为分类损失，以解决多标签草药毒性预测中的不平衡问题。ASL 通过对不同正负样本分配不同的惩罚，有效减轻标签不平衡的影响，从而增强模型对稀有但关键的毒性标签的关注。损失

3.3. Contrastive learning with dynamic coefficient

The above herb representations may not effectively capture toxicity-related similarities and differences among herbs, especially under setting where each herb can possess multiple toxic labels. Therefore, we propose a contrastive learning with dynamic coefficient strategy to update the herb representations. Canonical contrastive learning based methods aim to minimize the distance between instances of the same class while maximizing that between instances of different ones (Chen et al., 2020; Hadsell et al., 2006; Khosla et al., 2020). However, in multi-label herb toxicity prediction, two herbs often share only a subset of toxic labels, and each herb posses unique toxic labels. This multi-label phenomenon imposes greater complexity than the single-label setting, making it hard to simply treat two herbs as either “positive” or “negative” pairs.

To capture the intricate relationships between herbs with multiple toxic labels, we define a dynamic coefficient based on toxic label semantic similarity. Unlike binary-valued indicators in single-label settings, this coefficient provides a flexible way to numerically quantify the degree of overlapped labels between herb pairs, allowing the model to account for the complex relationships inherent in multi-label toxicity prediction tasks. Specifically, we measure the feature similarity between two herbs as:

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{||\mathbf{h}_i||_2\, ||\mathbf{h}_j||_2} \tag{9}$$

where $\cos(\ldots,\ldots)$ denotes the cosine similarity, $\mathbf{h}_i$ and $\mathbf{h}_j$ are the learned embeddings of the ith and jth herbs, respectively, and $||\cdot||_2$ represents the L_2 norm.

To further incorporate toxic label relationships, we define the label semantic similarity β_{ij} as the fraction of shared toxic labels between two herbs:

$$\beta_{ij} = \frac{\mathbf{y}_i^T \cdot \mathbf{y}_j}{\mathbf{y}_i^T \cdot \mathbf{y}_i} \tag{10}$$

where $\mathbf{y}_i \in \{0,1\}^L$ and $\mathbf{y}_j \in \{0,1\}^L$ are the multi-hot toxic label vectors for the ith and jth herbs, respectively, and L is the number of toxic labels. The numerator, $\mathbf{y}_i^T \cdot \mathbf{y}_j$, counts the number of shared toxic labels between the two herbs, while the denominator, $\mathbf{y}_i^T \cdot \mathbf{y}_i$, represents the total number of toxic labels for the ith herb. This ratio-based semantic similarity enables the model to distinguish not only completely identical or disjoint label sets in single-label setting, but also to quantify the partial similarity unique to multi-label scenario.

During subsequent experiments, we found that β may not sufficiently suitable to act as the dynamic coefficient for contrastive learning, as it does not provide a robust measure of relevance between two herbs. In particular, β can assign an excessively large coefficient to herb samples with low toxic label similarity. This issue can degrade the performance of herb toxicity prediction. As a result, we introduce a tunable hyperparameter t to adjust β to obtain a more appropriate dynamic coefficient δ:

$$\delta_{ij} = (\beta_{ij})^t \tag{11}$$

The contrastive learning loss with dynamic coefficient δ_{ij} for a herb pair is defined as:

$$\mathcal{L}_{ij}^{con} = -\delta_{ij} \log \frac{\exp\left(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau\right)}{\sum_{k=1,k\neq i}^{N} \exp\left(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau\right)} \tag{12}$$

where τ is the temperature hyperparameter used to scale the similarity scores, and N is the total number of herbs in the batch. This formulation allows the model to emphasize similar herb pairs with higher toxicity similarity while effectively learning representations for herbs with multiple related toxic labels. This formulation explicitly leverages the multi-label toxicity of herbs, since each herb pair’s contribution to the loss is scaled by the extent of their shared toxic labels.

For a given pair of herbs $(\mathbf{h}_i, \mathbf{h}_j)$, a higher semantic similarity β_{ij} results in a larger dynamic coefficient δ_{ij}, which in turn leads to a larger loss $\mathcal{L}_{ij}^{con}$. Consequently, the feature similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized toward a higher value. Conversely, if two samples do not share any labels ($\beta_{ij} = \delta_{ij} = 0$), the value $\mathcal{L}_{ij}^{con}$ becomes zero, and their cosine similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized to a lower value. The contrastive learning loss with dynamic coefficient for N herbs is computed as:

$$\mathcal{L}^{con} = \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \mathcal{L}_{ij}^{con} \tag{13}$$

3.4. Herb toxicity prediction based on weighted label fusion

For the ith herb, its representation $\mathbf{h}_i$ is passed through a three-layer MLP, followed by a sigmoid activation function, to compute the probability vector for all toxic labels:

$$\tilde{\mathbf{y}}_i^{mlp} = \text{sigmoid}(\text{MLP}(\mathbf{h}_i)) \tag{14}$$

where MLP(·) denotes a three-layer MLP. $\tilde{\mathbf{y}}_i^{mlp} \in \{0,1\}^L$ represents the predicted probability vector that the ith herb is associated with toxic label.

However, the above MLP may not fully utilize the valuable knowledge encoded in the training data. To further enhance the toxicity prediction, we introduce a weighted label fusion (WLF) strategy. For each target herb, we retrieve training herbs with similar representations. By virtue of contrastive learning, these neighbors are more likely to share meaningful toxicity patterns with the target herb. HerbToxNet aggregates the toxic labels of neighborhood herbs in a weighted manner, which complements the MLP’s output and contributes to more accurate toxicity prediction.

Let $\mathcal{D} = \{(\mathbf{h}_j, \mathbf{y}_j)\}_{j=1}^N$ denote the training set with N herbs. Given the ith target herb with representation $\mathbf{h}_i$, we retrieve its relevant neighbors from the training set by computing the cosine similarity between $\mathbf{h}_i$ and all $\mathbf{h}_j$ in $\mathcal{D}$. Neighbors whose similarity exceeds a predefined threshold σ are selected to form the neighbor set:

$$\mathcal{N}_\sigma = \{(\mathbf{h}_j, \mathbf{y}_j) \mid \cos(\mathbf{h}_i, \mathbf{h}_j) \geq \sigma\} \tag{15}$$

Instead of using K-nearest neighbors, our threshold-based strategy adaptively selects neighbors that are close to the target herb in the representation space.

The WLF-based prediction for the ith herb is obtained by aggregating the label vectors of the selected neighbors, weighted by their similarities with respect to the target herb:

$$\tilde{\mathbf{y}}_i^{wlf} = \sum_{(\mathbf{h}_j, \mathbf{y}_j) \in \mathcal{N}_\sigma} \mu_j \mathbf{y}_j \tag{16}$$

where the weight μ_j is defined as:

$$\mu_j = \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau')}{\sum_{(\mathbf{h}_k, \mathbf{y}_k) \in \mathcal{N}_\sigma} \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau')} \tag{17}$$

with τ' being a temperature hyperparameter that controls the sharpness of the weight distribution. In this way, neighbors with higher similarity to the ith herb are assigned with larger weights, which more significantly contribute to the WLF-based prediction than other neighbors.

To this end, we combine the MLP-based prediction $\tilde{\mathbf{y}}_i^{mlp}$ with WLF-based prediction $\tilde{\mathbf{y}}_i^{wlf}$ as:

$$\tilde{\mathbf{y}}_i = \alpha \tilde{\mathbf{y}}_i^{wlf} + (1-\alpha) \tilde{\mathbf{y}}_i^{mlp} \tag{18}$$

where the scalar parameter α balances the two sources of predictions.

We adopt the asymmetric loss (ASL) (Ridnik et al., 2021) as the classification loss to address the imbalanced nature of multi-label herb toxicity prediction. ASL effectively mitigates the impact of label imbalance by assigning different penalties to positive and negative samples, thereby enhancing the model’s focus on rare but critical toxic labels. The loss

函数 定义为 如下： 定义 为

$$\mathcal{L}^{asl} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{L}\Big[y_{ij}(1-\tilde{y}_{ij})^{\gamma^+}\log(\tilde{y}_{ij}) + (1-y_{ij})(p_{ij}^{m})^{\gamma^-}\log(1-p_{ij}^{m})\Big] \tag{19}$$

$$p_{ij}^{m} = \max\left(\tilde{y}_{ij}-m, 0\right) \tag{20}$$

其中 γ^+ 和 γ^- 分别是正样本和负样本的聚焦参数，$m \geq 0$ 是概率裕度。

HerbToxNet 的整体损失函数由 ASL 分类损失和对比学习损失组成，公式表述为：

$$\mathcal{L} = \mathcal{L}^{asl} + \lambda\mathcal{L}^{con} \tag{21}$$

where λ is a the 超参数 that 平衡 the 两个 损失。算法 1 summarizes the 伪 代码 of HerbToxNet

算法 1 HerbToxNet 的整体过程。输入：草药药理学 $\mathbf{p}_i$，功效 $\mathbf{e}_i$；毒性标签 $\mathbf{y}_i$；元路径集 $\mathcal{P}$；节点集 $\mathcal{V}$；边集 $\mathcal{E}$；模型参数 Θ；输出：预测的草药毒性标签 $\tilde{\mathbf{y}}$；1：▷ 异构图构建 2: 构建节点集 $\mathcal{V}_h$ (草药)，$\mathcal{V}_c$ (成分)，$\mathcal{V}_t$ (靶点) 和边集 $\mathcal{E}_{hc}$, $\mathcal{E}_{ht}$, $\mathcal{E}_{ct}$, $\mathcal{E}_{hh}$; 3: 对于每个草药i，通过公式 (1) 初始化节点特征 $\mathbf{f}_i$；4: 组装异构图$\mathcal{G} = (\mathcal{V}, \mathcal{E})$；5：▷ 基于 HAN 的草药表示学习 6: 对于每个元路径 $p_k \in \mathcal{P}$ do 7: 从$\mathcal{G}$中提取元路径子图 $\mathcal{G}_{p_k}$；8: 对于每个节点i do 9: 通过公式 (2) 计算注意力分数 $e_{ij}^{\mathcal{G}_{p_k}}$ 并通过公式 (3) 进行归一化；10: 通过公式 (4) 获取邻居聚合特征 $\mathbf{f}_i^{\mathcal{G}_{p_k}}$；11: 通过公式 (5) 获取多头特征表示 $\mathbf{z}_i^{\mathcal{G}_{p_k}}$；12: end for 13: end for 14: 通过公式 (6) 计算元路径注意力 $w_{\mathcal{G}_{p_k}}$ 并通过公式 (7) 进行归一化；15: 通过公式 (8) 融合元路径嵌入以获取 $\mathbf{h}_i$；16: ▷ 具有动态系数的对比学习 17: 对于每个草药对 (i, j) do 18: 通过公式 (9) 计算特征相似度；19: 通过公式 (10) 和公式 (11) 计算语义相似度 β_{ij} 和动态系数 δ_{ij}；20: end for 21: 通过公式 (13) 计算总对比损失 $\mathcal{L}^{con}$；22: ▷ 草药毒性预测 23: 对于每个草药i do 24: MLP 通过公式 (14) 预测 $\tilde{\mathbf{y}}_i^{mlp}$；25: 通过公式 (18) 将$\tilde{\mathbf{y}}_i^{mlp}$ 和$\tilde{\mathbf{y}}_i^{wlf}$ 加权融合到$\tilde{\mathbf{y}}_i$ 中；26: end for 27: 通过公式 (19) 和公式 (20) 计算非对称损失 $\mathcal{L}^{asl}$；28: 通过公式 (21) 计算总损失 $\mathcal{L}$；29: 通过反向传播和 Adam 更新模型参数Θ；30: return 草药预测的毒性标签$\tilde{\mathbf{y}}$；

4. 实验

4.1. 实验 设置

4.1.1. 数据集

我们 收集了 药理 性质、 功效、 和 交互 数据 用于 草药 从 已建立 数据库、 包括 ChP (2020) (徐等人，2021年)，HERB (方等人，2021年)，SymMap v2 (吴等人，2019b)，TCMBank (李等人，2023年)，和HIT 2.0 (严等人，2022b)。毒性标签是从主要文献和数据库来源如Google Scholar、CNKI、VIP、万方、PubMed和Web of Science获得的，并分为五类：心脏毒性、肾毒性、肝毒性、全身毒性和其他毒性。表1概述了数据来源。

我们的数据集的整理流程在补充文件的I节中有报道。然后，我们将已知草药的毒性标签作为正样本，其他作为该草药的负样本。我们保留15% 的草药作为测试集，而其余草药用于五折交叉验证。此过程独立重复十次以确保鲁棒性。为确保完整性，在划分过程中，与草药相关的所有交互都被分组到同一子集中。

4.1.2. 评估 指标

为了评估HerbToxNet和基线方法的性能，我们采用了四个广泛使用的评估指标（Qiu等人，2023年）：宏平均F1分数（Macro-F1）、微平均F1分数（Micro- F1）、接收者操作特征曲线下面积平均值（AvgAUC）和一错误率（OneError）。Macro-F1独立地为每个标签计算F1分数，然后取平均值，反映所有标签的整体性能。Micro-F1汇总所有标签的贡献来计算平均F1分数，反映所有样本的整体性能。AvgAUC为每个标签计算AUC，然后取平均值，更高的值意味着毒性标签和非毒性标签之间的可分离性更好。OneError计算预测的毒性标签中排名第一的标签不在真实标签中的数量；更低的值表明模型更有可能将真实标签排在第一位。为了与其他指标保持一致，我们使用1 - OneError。

4.1.3. 基线 方法

为证明我们提出的HerbToxNet的有效性，我们选择了几种代表性方法作为基线，这些方法可以归纳为三类：草药毒性预测方法、化学药物毒性预测方法以及生物医学领域的多标签预测方法。(i) 草药毒性预测方法：ZF- LightGBM（余等人，2025）在正清风宁诱导的肝损伤数据集上比较了十种机器学习模型，发现LightGBM表现最佳；QSAR-ANN和QSAR-SVM（孙等人，2019）使用草药肾毒性数据构建QSAR模型，分别应用人工神经网络和支持向量机算法预测肾毒性；EI-MS-RF（贾等人，2024）通过将草药毒性与电子电离质谱数据的先进分析描述符相关联，结合可解释机器学习模型进行毒性预测。(ii) 化学药物毒性预测方法：DeepDILI（李等人，2020）结合传统机器学习算法生成的模型级表示与基于Mold2描述符的深度学习框架，预测药物引起的肝损伤；R-E-GA（严等人，2022a）将特征空间中高维分子的融合向量进行旋转，以找到具有更好预测性能的特征子集，然后将基于Adaboost的集成学习方法集成到R-E-GA中以提高预测精度；TIRPnet（魏等人，2024）旨在从丰富的中医药自发报告中解析知识，从而对个体成分的潜在不良药物反应（ADRs）进行预测。(iii) 多标签预测方法：ML-PRDF（龚等人，2023）首先利用PCC-MLRF（龚等人，2023）进行特征选择，然后将精心选择的特征输入到多标签深度森林模型中进行区分和预测分析；MulGRaL（Pham等人，2022）构建了一个与预测结果相关的异构信息网络，随后结合多个多标签分类器以提高预测的准确性和可靠性；MLC-CNN（葛等人，2021）根据标签之间的相互关系对多标签进行分组，并利用神经网络模型进行预测，

function is defined as follows:

$$\mathcal{L}^{asl} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{L}\Big[y_{ij}(1-\tilde{y}_{ij})^{\gamma^+}\log(\tilde{y}_{ij}) + (1-y_{ij})(p_{ij}^{m})^{\gamma^-}\log(1-p_{ij}^{m})\Big] \tag{19}$$

$$p_{ij}^{m} = \max\left(\tilde{y}_{ij}-m, 0\right) \tag{20}$$

where γ^+ and γ^- are focusing parameters for positive and negative samples, respectively, and $m \geq 0$ is the probability margin.

The overall loss function of HerbToxNet consists of the ASL classification loss and the contrastive learning loss, formulated as:

$$\mathcal{L} = \mathcal{L}^{asl} + \lambda\mathcal{L}^{con} \tag{21}$$

where λ is a the hyperparameter that balances the two losses. Algorithm 1 summarizes the pseudo code of HerbToxNet

Algorithm 1 The overall process of HerbToxNet.

Input: herb pharmacology $\mathbf{p}_i$, efficacy $\mathbf{e}_i$; toxic labels $\mathbf{y}_i$; meta-path set $\mathcal{P}$; node sets $\mathcal{V}$; edge sets $\mathcal{E}$; model parameters Θ;
Output: Predicted herb toxic label $\tilde{\mathbf{y}}$;

```
▷ Heterogeneous Graph Construction
Build node set V_h (herb), V_c (ingredient), V_t (target) and edge set E_hc, E_ht, E_ct, E_hh;
For each herb i, initialize node features f_i via Eq. (1);
Assemble the heterogeneous graph G = (V, E);
▷ Herb Representation Learning by HAN
for each meta-path p_k ∈ P do
  Extract meta-path subgraph G_{p_k} from G;
  for each node i do
    Compute attention score e_ij^{G_{p_k}} via Eq. (2) and normalize it via Eq. (3);
    Get neighbor-aggregated features f_i^{G_{p_k}} via Eq. (4);
    Obtain multi-head feature representation z_i^{G_{p_k}} via Eq. (5);
  end for
end for
Compute meta-path attention w_{G_{p_k}} via Eq. (6) and normalize it via Eq. (7);
Fuse meta-path embeddings to get h_i via Eq. (8);
▷ Contrastive Learning with Dynamic Coefficient
for each herb pair (i, j) do
  Compute feature similarity via Eq. (9);
  Compute semantic similarity β_ij and dynamic coefficient δ_ij via Eq. (10) and Eq. (11);
end for
Compute total contrastive loss L^con via Eq. (13);
▷ Herb Toxicity Prediction
for each herb i do
  MLP predicts ỹ_i^mlp via Eq. (14);
  Weighted fuse ỹ_i^mlp and ỹ_i^wlf into ỹ_i via Eq. (18);
end for
Compute asymmetric loss L^asl via Eq. (19) and Eq. (20);
Compute total loss L via Eq. (21);
Update model parameters Θ by back propagation and Adam;
return Predicted toxic labels of herb ỹ;
```

4. Experiment

4.1. Experimental setup

4.1.1. Dataset

We collected pharmacological properties, efficacy, and interaction data for herbs from established databases, including ChP (2020) (Xu et al., 2021), HERB (Fang et al., 2021), SymMap v2 (Wu et al., 2019b), TCMBank (Lv et al., 2023), and HIT 2.0 (Yan et al., 2022b). Toxicity labels were obtained from major literature and database sources such as Google Scholar, CNKI, VIP, WanFang, PubMed, and Web of Science, and were grouped into five categories: cardiotoxicity, nephrotoxicity, hepatotoxicity, systemic toxicity, and other toxicity. Table 1 gives an overview of the data sources.

The curation pipeline of our dataset is reported in Section I of Supplementary file. Then, we take known toxic labels of a herb as positive samples and others as negative ones of this herb. We keep 15% of the herbs as the testing set, while the remaining herbs are used for five-fold cross-validation. This process is independently repeated ten times to ensure robustness. To ensure the integrity, all interactions associated with a herb are grouped into the same subset during the partition process.

4.1.2. Evaluation metrics

To evaluate the performance of HerbToxNet and the baseline methods, we employed four widely used evaluation metrics (Qiu et al., 2023): Macro-averaged F1 Score (Macro-F1), Micro-averaged F1 Score (Micro-F1), Average Area Under the receiver operating characteristic Curve (AvgAUC), and One Error (OneError). Macro-F1 independently calculates the F1 score for each label and then takes the average, reflecting the overall performance across all labels. Micro-F1 aggregates the contributions of all labels to compute the average F1 score, reflecting overall performance on all samples. AvgAUC computes the AUC for each label and then takes the average, a higher value means a better separability between toxic and non-toxic labels. OneError counts the number of the predicted toxic label with top-rank is not among the true labels; a lower values indicate that the model is more likely to rank a true label first. For consistency with other metrics, we use *1 - OneError* instead.

4.1.3. Baseline methods

To prove the effectiveness of our proposed HerbToxNet, several representative methods were selected as the baseline, which can be summarized into three categories: herb toxicity prediction methods, chemical drug toxicity prediction methods, and multi-label prediction methods in the field of biomedicine. (i) Herb toxicity prediction methods: ZF-LightGBM (Yu et al., 2025) compares ten machine learning models on *Zhengqing Fengtongning*-induced liver injury dataset and finds LightGBM performing the best; QSAR-ANN and QSAR-SVM (Sun et al., 2019) uses herb nephrotoxicity data to construct QSAR model, and apply ANN and SVM algorithms to predict nephrotoxicity, respectively; EI-MS-RF (Jia et al., 2024) makes toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. (ii) Chemical drug toxicity prediction methods: DeepDILI (Li et al., 2020) combines model-level representations generated by traditional machine learning algorithms with a deep learning framework based on Mold2 descriptors to predict drug-induced liver injury; R-E-GA (Yan et al., 2022a) rotates the fusion vector represented by a high-dimensional molecule in the feature space to find a subset of features with better predictive performance, and then integrates an Adaboost-based ensemble learning method into R-E-GA to improve prediction accuracy; TIRPnet (Wei et al., 2024) is designed to decipher knowledge from abundant of TCM spontaneous reports, enabling the prediction of potential adverse drug reactions (ADRs) for individual ingredient. (iii) multi-label prediction methods: ML-PRDF (Gong et al., 2023) initially utilizes PCC-MLRF (Gong et al., 2023) for feature selection, and then inputs the meticulously selected features into the multi-label deep forest model to conduct differentiation and predictive analysis; MulGRaL (Pham et al., 2022) constructs a heterogeneous information network relevant to the prediction outcomes, subsequently combines multiple multi-label classifiers to enhance the accuracy and reliability of prediction; MLC-CNN (Ge et al., 2021) involves grouping multi-labels according to their inter-label correlations and utilizes neural network models to perform predictions for

表 1
Details of the collected data sources.

	Type	Size	来源
Herb	Pharmaco logical Property	252×33	Chp(2020), HERB, SymMap v2, TCM-Bank,HIT 2.0
	功效	252×331	
	Herb-Target	34988	
	Herb-成分	3275	
	成分-靶点	5932	
毒性	心脏毒性	74	Google Scholar CNKI VIP WanFang , , , PubMed WOS
	肾毒性	101	
	肝毒性	114	
	系统ic Toxicity	167	
	其他oxicity	150	

Note: The size of 'Efficacy' is 252×331 indicating that the number of herbs is 252 and the , number of efficacy is 331. 'Pharmacological Property' is represented in the same way as '功效' 。

每个 组中， 有效 利用 了 不同 特 征 对 区 分 标<style id='23'> 签。

由于前两类方法最初是为二元分类任务开发的，我们将它们的最终二元输出修改为适合我们任务的多标签预测。此外，由于我们的数据集中没有EI-MS数据，我们用草药药理性质和功效数据替代EI-MS描述符，并实施了与原始工作中描述的相同的EI-MS-RF流程。对于每种基线方法，我们使用推荐参数作为起点，并根据我们的数据集进行必要的微调。所有基线方法以及我们的HerbToxNet的详细参数，以及HerbToxNet的综合参数敏感性分析，均在补充文件的第三节中呈现。

4.2. 结果 比较 与 现有 方案

表 2 总结了 HerbToxNet 和 基线方法 的性能 。HerbToxNet 在 大多数 评估 指标中 实现了 最佳的 草药毒性 预测 性能 。其他 重要 观察结果 如下:
(i) HerbToxNet 显著优于其他草药毒性预测方法，包括 ZF-LightGBM、QSAR-SVM、EI-MS-RF 和 QSAR-人工神经网络，主要原因是它们无法有效捕捉草药复杂的成分-靶点异构网络特征。这种缺陷限制了它们的泛化能力，这在处理多标签毒性预测任务时尤为明显。此外，它们通常预测单个毒性标签，并且无法有效建模毒性标签之间的复杂关联，从而限制了它们在多标签毒性预测中的性能。相比之下，HerbToxNet 在复杂的多标签毒性预测任务中表现更优，在大多数评估指标上显著领先。其优势如下。首先，HerbToxNet 构建了草药-成分-靶点异构网络，并利用元路径提取草药的属性和额外信息，实现对草药复杂特征的精确建模。其次，它引入了具有动态系数的对比学习策略，使具有相似毒性标签的草药对表示更接近，而不同毒性标签的草药对表示被推远，增强了模型对多标签任务的适应性。第三，HerbToxNet 引入了加权标签融合策略，有效利用相似草药的毒性标签，从而进一步提高预测性能并减轻稀疏草药数据的影响。(ii) HerbToxNet 还显著优于药物毒性预测方法，包括 DeepDILI、R-E-GA 和 TIRPnet。这些方法在草药的多标签毒性预测任务中表现不佳，因为化学药物通常具有明确的分子结构和指纹，为预测提供了丰富和详细的信息。相比之下，草药由多成分组成，具有低维度的药理学数据，而独热编码用于表示这些成分不足以捕捉其复杂特征。此外，它们可能无法有效捕捉和利用与草药相关的毒性标签之间的关联。HerbToxNet 通过利用异构网络建模、具有动态系数的对比学习和加权标签融合来解决这些挑战。这些创新使 HerbToxNet 能够有效捕捉草药的复杂特征，并显著提升其在多标签毒性预测任务中的性能。

(iii) 多标签分类方法在预测草药的毒性标签方面显示出一定的优势，与其他两种方法相比。这是因为它们可以同时考虑多个毒性标签，这天然适合多标签草药毒性预测。尽管这些方法在某些场景下可以提供具有竞争力的结果，但它们可能在挖掘有限草药数据的内在信息和复杂特征方面受到限制。因此，如表2所示，这些方法再次输给HerbToxNet。HerbToxNet的优越性能主要归因于其有效的建模

表 2
Performance comparison of HerbToxNet and baseline methods in predicting herb toxi-城市。

	Macro F1	微观 F1	AvgAUC	1 - OneError
ZF-LightGBM	0.2371±0.0413	0.4127±0.0420	0.5163±0.0307	0.2535±0.0487
QSAR-ANN	0.4195±0.0460	0.4510±0.0429	0.5105±0.0343	0.1936±0.0536
QSAR-SVM	0.1619±0.0235	0.3842±0.0422	0.5132±0.0422	0.2582±0.0520
EI-MS-RF	0.3432±0.0384	0.4437±0.0304	0.5120±0.0370	0.2242±0.0482
DeepDILI	0.3035±0.0956	0.4756±0.0842	0.5166±0.0489	0.5329±0.0702
R-E-GA	0.2894±0.0470	0.4220±0.0465	0.5092±0.0352	0.2313±0.0417
TIRPnet	0.4089±0.0420	0.4265±0.0468	0.5058±0.0384	0.2111±0.0608
ML-PRDF	0.4059±0.0327	0.4508±0.0439	0.5294±0.0548	0.4711±0.0698
MulGRaL	0.4135±0.0421	0.5576±0.0325	0.5606±0.0209	0.5195±0.0611
MLC-CNN	0.4966±0.0653	0.5086±0.0508	0.5525±0.0349	0.5115±0.0676
	0.5652± 0.0161	0.6247±0.0172	0.6003± 0.0141	0.5587±0.0485

注意: The best results are highlighted in bold font with statistical significance checked , % by student t-test at 95 level.

Table 1
Details of the collected data sources.

	Type	Size	Sources
Herb	Pharmacological Property	252×33	Chp (2020), HERB, SymMap v2, TCM-Bank, HIT 2.0
	Efficacy	252×331	
	Herb-Target	34988	
	Herb-Ingredient	3275	
	Ingredient-Target	5932	
Toxicity	Cardiotoxicity	74	Google Scholar, CNKI, VIP, WanFang, PubMed, WOS
	Nephrotoxicity	101	
	Hepatotoxicity	114	
	Systemic Toxicity	167	
	Other Toxicity	150	

Note: The size of 'Efficacy' is 252×331, indicating that the number of herbs is 252 and the number of efficacy is 331. 'Pharmacological Property' is represented in the same way as 'Efficacy'.

each group, effectively leveraging the impact of diverse features on distinct labels.

Since the first two categories of methods were originally developed for binary classification tasks, we modify their final binary outputs to produce multi-label predictions suitable for our task. In addition, as EI-MS data are not available in our dataset, we substitute the EI-MS descriptors with herb pharmacological property and efficacy data, and implement the same EI-MS-RF pipeline as described in the original work. For each baseline method, we used the recommended parameters as a starting point and fine-tuned them as necessary for our dataset. Detailed parameters for all baselines and our HerbToxNet, as well as a comprehensive parameter sensitivity analysis of HerbToxNet, are presented in Section III of Supplementary file.

4.2. Results comparison with existing solutions

Table 2 summarizes the performance of HerbToxNet and baselines. HerbToxNet achieves the best herb toxicity prediction performance in most evaluation metrics. Other important observations are:
(i) HerbToxNet significantly outperforms other herb toxicity prediction methods, including ZF-LightGBM, QSAR-SVM, EI-MS-RF, and QSAR-ANN, primarily due to they inability to effectively capture the complex multi-ingredient and multi-target characteristics of herbs. This defect limits their generalization ability, which is particularly evident when handling multi-label toxicity prediction tasks. Moreover, they typically predict single toxic labels and fail to effectively model the intricate correlations between toxic labels, thereby restricting their performance in multi-label toxicity prediction. In contrast, HerbToxNet achieves better performance in complex multi-label toxicity prediction tasks, significantly leading in most evaluation metrics. Its advantages are as follows. First, HerbToxNet constructs a herb-ingredient-target heterogeneous network and utilizes meta-path to extract properties and relational information of herbs, achieving precise modeling of complex characteristics of herbs. Second, it introduces a contrastive learning with dynamic coefficient strategy that brings closer the representations of herb pairs with similar toxic labels while pushing apart those with different toxic labels, enhancing the model's adaptability to multi-label tasks. Third, HerbToxNet introduces a weighted label fusion strategy to effectively utilize the toxic labels of similar herbs, thus further improving the prediction performance and alleviating the impact of scarce herb data.
(ii) HerbToxNet also significantly outperforms drug toxicity prediction methods, including DeepDILI, R-E-GA, and TIRPnet. These methods perform poorly in the multi-label toxicity prediction task for herb, because chemical drugs typically possess well-defined molecular structures and fingerprints, which provide abundant and detailed information for prediction. In contrast, a herb comprises of multi-ingredients with lower-dimensional pharmacological data, and the one-hot encoding to represent these ingredients is insufficient to capture their complex characteristics. Moreover, they may not effectively capture and leverage the correlations between toxic labels associated with herbs. HerbToxNet addresses these challenges by leveraging heterogeneous network modeling, contrastive learning with dynamic coefficient, and weighted label fusion. These innovations enable HerbToxNet to effectively capture the intricate features of herbs and significantly enhance its performance in multi-label toxicity prediction tasks.
(iii) Multi-label classification-based methods show certain advantages in predicting the toxic labels of herbs, compared with the other two kinds of methods. That is because they can consider multiple toxicity labels simultaneously, which is naturally suited for multi-label herb toxicity prediction. Although these methods can provide competitive results in some scenarios, they may be limited in mining the intrinsic information and complex characteristics of limited herb data. Therefore, as shown in Table 2, these methods again lose to HerbToxNet. The superior performance of HerbToxNet is mainly attributed to its effective modeling

Table 2
Performance comparison of HerbToxNet and baseline methods in predicting herb toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.2371±0.0413	0.4127±0.0420	0.5163±0.0307	0.2535±0.0487
QSAR-ANN	0.4195±0.0460	0.4510±0.0429	0.5105±0.0343	0.1936±0.0536
QSAR-SVM	0.1619±0.0235	0.3842±0.0422	0.5132±0.0422	0.2582±0.0520
EI-MS-RF	0.3432±0.0384	0.4437±0.0304	0.5120±0.0370	0.2242±0.0482
DeepDILI	0.3035±0.0956	0.4756±0.0842	0.5166±0.0489	0.5329±0.0702
R-E-GA	0.2894±0.0470	0.4220±0.0465	0.5092±0.0352	0.2313±0.0417
TIRPnet	0.4089±0.0420	0.4265±0.0468	0.5058±0.0384	0.2111±0.0608
ML-PRDF	0.4059±0.0327	0.4508±0.0439	0.5294±0.0548	0.4711±0.0698
MulGRaL	0.4135±0.0421	0.5576±0.0325	0.5606±0.0209	0.5195±0.0611
MLC-CNN	0.4966±0.0653	0.5086±0.0508	0.5525±0.0349	0.5115±0.0676
	0.5652± 0.0161	0.6247±0.0172	0.6003± 0.0141	0.5587±0.0485

Note: The best results are highlighted in bold font, with statistical significance checked by student t-test at 95 % level.

表 3
~~Comparison of HerbToxNet with other methods on hepatotoxicity and nephrotoxicity prediction.~~

	H肝毒性 Pr预测			Nephr毒性 Pr预测		
	AUROC	AUPRC	F1	AUROC	AUPRC	F1
ZF-Li ghtGBM	0.5610±0.0803	0.5369±0.0944	0.6323±0.0565	0.5682±0.0554	0.4668±0.0723	0.5881±0.0577
QSAR-ANN	0.5543±0.0825	0.5409±0.1038	0.6427±0.0565	0.5571±0.0738	0.4847±0.0798	0.5973±0.0609
QSAR-SV M	0.5675±0.1038	0.5753±0.1197	0.6295±0.0574	0.5695±0.0649	0.4913±0.0695	0.5848±0.0585
EI-MS-RF	0.5624±0.0765	0.5304±0.0984	0.6435±0.0594	0.543 1±0.0743	0.452 0±0.078 2	0.6023±0.0566
DeepDILI	0.5486±0.0763	0.5658±0.1005	0.6272±0.0580	0.5799±0.0744	0.4989±0.0782	0.5949±0.0550
R-E-GA	0.534 2 ±0.0786	0.5437±0.1011	0.6218±0.0574	0.5627±0.0804	0.5053±0.0939	0.5728±0.0578
TIRPnet	0.5269±0.0824	0.4959±0.0907	0.5084±0.0918	0.4949±0.0722	0.4279±0.0735	0.4630±0.0761
HerbToxN et	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293

注意： The best results are highlighted in bold font, with statistical significance checked by student t-test at 95 % level.

of 草药, 成分 和 靶点 通过 异构 网络, 对比 学习 和 加权 标签 融合。

4.3. 肝毒性 和 肾毒性 预测

HerbToxNet 旨在进行多标签毒性预测，而大多数方法针对的是草药的单标签毒性预测（贾等人，2024；孙等人，2019；余等人，2025）和药物的单标签毒性预测（李等人，2020；魏等人，2024；严等人，2022a），它们基于单类型毒性的二元任务。为了评估 HerbToxNet 是否可以适应这些二元任务，我们从数据集中分离出肝毒性和肾毒性草药，将它们分为肝毒性数据集和肾毒性数据集，并在两个数据集中评估我们的方法。其中，肝毒性数据集包含 114 种草药，肾毒性数据集包含 101 种草药。由于这是一个二元任务，因此使用 AUROC、AUPRC 和 F1 进行评估。

如表3所示，尽管HerbToxNet最初是为多标签毒性预测而设计的，但在应用于专注于单个毒性类型的二元分类任务时，它表现出良好的适应性和鲁棒性。在肝毒性预测任务中，HerbToxNet在所有评估指标中均取得了最佳性能，超越了所有基线模型。与Deep-DILI、基于QSAR的方法和EI-MS-RF等现有方法相比，HerbToxNet实现了更高的AUPRC值，这表明它在类别不平衡设置下识别阳性样本的能力更强。类似地，在肾毒性预测任务中，HerbToxNet取得了最高的AUROC、AUPRC和F1分数，这证明它在识别肾毒性草药方面具有强大的区分能力。值得注意的是，HerbToxNet在两种毒性类型中均获得了持续更优的结果，验证了它在不同毒理学终点上的稳定适应性和学习能力。

这些 结果 表明 HerbToxNet 不仅 在 建模 复杂 多标签 毒性 关系 方面 有效， 而且 能够 提取 关键 特征 并 评估 毒性 风险， 适用于 二分类 任务。

4.4. 药物 毒性 预测

HerbToxNet不限于<style id='1'> </style>草药毒性预测<style id='3'> </style>，<style id='5'> </style>还可以扩展到<style id='9'> </style>化学药物<style id='11'> </style>。为了研究<style id='13'> </style>其泛化能力<style id='15'> </style>，<style id='17'> </style>我们在UniTox数据集（Silberg等人，2024年）上<style id='19'> </style>进行评估<style id='21'> </style>，<style id='23'> </style>该数据集包含<style id='25'> </style>1349种化学化合物<style id='27'> </style>，<style id='29'> </style>并带有<style id='31'> </style>八个毒性标签<style id='33'> </style>。鉴于<style id='35'> </style>化学药物和草药在结构和语义上的差异<style id='37'> </style>，<style id='39'> </style>我们移除了异构图和HAN<style id='41'> </style>，<style id='43'> </style>直接使用每个药物的SMILES表示作为输入<style id='45'> </style>。从SMILES字符串中提取Morgan指纹以表示分子特征<style id='47'> </style>，<style id='49'> </style>这些特征随后与对比学习模块和加权标签融合策略相结合<style id='51'> </style>，<style id='53'> </style>用于<style id='55'> </style>化学药物的毒性预测<style id='57'> </style>。在基线方法中<style id='59'> </style>，<style id='61'> </style>MulFRaL被排除在比较之外<style id='63'> </style>，<style id='65'> </style>因为它高度依赖异构图结构<style id='67'> </style>，<style id='69'> </style>在本设置中不适用<style id='71'> </style>。所有其他基线方法保持不变<style id='73'> </style>。实验设置和评估指标与草药毒性预测任务中使用的相同。

表4总结了HerbToxNet和基线方法在预测药物毒性的性能。HerbToxNet在大多数评估指标上的结果优于基线方法。它在化学药物毒性预测任务中再次取得了优异的性能，这主要归功于其在特征表示和标签预测方面的方法论优势。尽管在此设置中排除了异构图和HAN，HerbToxNet利用对比学习模块来增强药物表示的区分性。具体来说，对比学习策略鼓励具有相似毒性特征的药物表示在特征空间中更接近，这有助于区分不同的毒性标签。Morgan指纹的使用提供了一种紧凑且广泛适用的分子表示，这使得模型即使在缺乏特定于中医药的复杂语义上下文的情况下也能保持高预测效能。此外，WLF结合了来自多层感知器（MLP）的预测以及从相似草药聚合的预测，有效缓解了训练数据有限的问题。这些结果不仅验证了HerbToxNet的泛化能力，而且

表 4
Performance comparison of HerbToxNet and baseline methods in predicting drug tox-性。

	Macro F1	微观 F1	AvgAUC	1 - OneError
ZF-Li ghtGBM	0.6107±0.0169	0.6961±0.011 4	0.6973±0.0139	0.3610±0.022 6
QSAR-ANN	0.6196±0.0155	0.7160±0.0128	0.7059±0.0163	0.3250±0.0223
QSAR-SV M	0.5241±0.0212	0.7085±0.0109	0.7038±0.0172	0.3424±0.0281
EI-MS-RF	0.5696±0.0155	0.7129±0.0110	0.7038±0.0148	0.3689±0.0245
DeepDILI	0.5929±0.0174	0.7184±0.0132	0.7055±0.0173	0.8638±0.0234
R-E-GA	0.5382±0.0360	0.6538±0.0303	0.5998±0.0055	0.3714±0.0186
TIRPnet	0.5861±0.0383	0.6905±0.0162	0.6618±0.0226	0.3658±0.0563
ML-PRDF	0.6107±0.0159	0.7080±0.0117	0.6260±0.0129	0.8504±0.0149
MLC-CNN	0.5091±0.0682	0.6648±0.0301	0.6179±0.0265	0.8718±0.0186
HerbToxN et	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164

注意:
The best results are highlighted in bold font with statistical significance checked , % by student t-test at 95 level.

Table 3
Comparison of HerbToxNet with other methods on hepatotoxicity and nephrotoxicity prediction.

	Hepatotoxicity Prediction			Nephrotoxicity Prediction		
	AUROC	AUPRC	F1	AUROC	AUPRC	F1
ZF-LightGBM	0.5610±0.0803	0.5369±0.0944	0.6323±0.0565	0.5682±0.0554	0.4668±0.0723	0.5881±0.0577
QSAR-ANN	0.5543±0.0825	0.5409±0.1038	0.6427±0.0565	0.5571±0.0738	0.4847±0.0798	0.5973±0.0609
QSAR-SVM	0.5675±0.1038	0.5753±0.1197	0.6295±0.0574	0.5695±0.0649	0.4913±0.0695	0.5848±0.0585
EI-MS-RF	0.5624±0.0765	0.5304±0.0984	0.6435±0.0594	0.5431±0.0743	0.4520±0.0782	0.6023±0.0566
DeepDILI	0.5486±0.0763	0.5658±0.1005	0.6272±0.0580	0.5799±0.0744	0.4989±0.0782	0.5949±0.0550
R-E-GA	0.5342±0.0786	0.5437±0.1011	0.6218±0.0574	0.5627±0.0804	0.5053±0.0939	0.5728±0.0578
TIRPnet	0.5269±0.0824	0.4959±0.0907	0.5084±0.0918	0.4949±0.0722	0.4279±0.0735	0.4630±0.0761
HerbToxNet	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293

Note: The best results are highlighted in bold font, with statistical significance checked by student t-test at 95 % level.

of herbs, ingredients and targets by heterogeneous network, contrastive learning and weighted label fusion.

4.3. Hepatotoxicity and nephrotoxicity prediction

HerbToxNet is designed for multi-label toxicity prediction, while most methods target at single-label toxicity prediction of herbs (Jia et al., 2024; Sun et al., 2019; Yu et al., 2025) and of drugs (Li et al., 2020; Wei et al., 2024; Yan et al., 2022a), they build on binary tasks for single type toxicity. To assess whether HerbToxNet can be adapted to these binary tasks, we isolated hepatotoxicity and nephrotoxicity herbs from the dataset, divided them as hepatotoxicity and nephrotoxicity datasets and evaluated our approach in both datasets. Among them, the hepatotoxicity dataset contains 114 herbs and the nephrotoxicity dataset contains 101 herbs. Since this is a binary task, AUROC, AUPRC, and F1 are used for evaluation.

As shown in Table 3, although HerbToxNet is originally designed for multi-label toxicity prediction, it demonstrates good adaptability and robustness when applied to binary classification tasks focused on individual toxicity types. In the hepatotoxicity prediction task, HerbToxNet achieves the best performance in all evaluation metrics, outperforming all baseline models. Compared with existing methods such as DeepDILI, QSAR-based approaches, and EI-MS-RF, HerbToxNet achieves higher AUPRC values, suggesting a better capability to identify positive samples under class-imbalanced settings. Similarly, HerbToxNet holds competitive performance in the nephrotoxicity prediction task, achieving the highest AUROC, AUPRC, and F1 scores, which demonstrates its strong discriminative ability in identifying nephrotoxic herbs. Notably, HerbToxNet obtains consistently better results across both toxicity types, validating its stable adaptability and learning capacity across different toxicological endpoints.

These results underscore that HerbToxNet is not only effective in modeling complex multi-label toxicity relationships, but also capable of extracting critical features and assessing toxicity risk in binary classification tasks.

4.4. Drug toxicity prediction

HerbToxNet is not limited to herb toxicity prediction, it can also be extended to chemical drugs. To study its generalizability, we evaluate it on the UniTox dataset (Silberg et al., 2024), which contains 1349 chemical compounds annotated with eight toxic labels. Given the structural and semantic differences between chemical drugs and herbs, we remove the heterogeneous graph and HAN, and directly use the SMILES representation of each drug as input. From the SMILES strings, we extract Morgan fingerprints to represent molecular features. These features are then integrated with the contrastive learning module and the weighted label fusion strategy to perform toxicity prediction for chemical drugs. Among the baseline methods, MulFRaL was excluded from comparison as it relies heavily on heterogeneous graph structures, which are not applicable in this setting. All other baseline methods keep unchanged. The experimental setup and evaluation metrics are the same as those used in the herb toxicity prediction task.

Table 4 summarizes the performance of HerbToxNet and baselines on predicting drug toxicity. The results of HerbToxNet are better than those of baseline methods in most evaluation metrics. It again achieves superior performance in the toxicity prediction task for chemical drugs, primarily due to its methodological strengths in both feature representation and label prediction. Although the heterogeneous graph and HAN are excluded in this setting, HerbToxNet leverages the contrastive learning module to enhance the discriminability of drug representations. Specifically, the contrastive learning strategy encourages representations of drugs with similar toxicity profiles to be closer in the feature space, this helps to distinguish different toxic labels. The use of Morgan fingerprints offers a compact and widely applicable molecular representation, which enables the model to maintain high prediction efficacy even in the absence of complex semantic contexts specific to TCM. Moreover, WLF combines the predictions from MLP and those aggregated from similar herbs, effectively alleviating the issue of limited training data. These results not only validate the generalizability of HerbToxNet, but

Table 4
Performance comparison of HerbToxNet and baseline methods in predicting drug toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.6107±0.0169	0.6961±0.0114	0.6973±0.0139	0.3610±0.0226
QSAR-ANN	0.6196±0.0155	0.7160±0.0128	0.7059±0.0163	0.3250±0.0223
QSAR-SVM	0.5241±0.0212	0.7085±0.0109	0.7038±0.0172	0.3424±0.0281
EI-MS-RF	0.5696±0.0155	0.7129±0.0110	0.7038±0.0148	0.3689±0.0245
DeepDILI	0.5929±0.0174	0.7184±0.0132	0.7055±0.0173	0.8638±0.0234
R-E-GA	0.5382±0.0360	0.6538±0.0303	0.5998±0.0055	0.3714±0.0186
TIRPnet	0.5861±0.0383	0.6905±0.0162	0.6618±0.0226	0.3658±0.0563
ML-PRDF	0.6107±0.0159	0.7080±0.0117	0.6260±0.0129	0.8504±0.0149
MLC-CNN	0.5091±0.0682	0.6648±0.0301	0.6179±0.0265	0.8718±0.0186
HerbToxNet	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164

Note: The best results are highlighted in bold font, with statistical significance checked by student t-test at 95 % level.

也 突出了 其 核心 设计 在不同 毒性预测 场景下的 方法论适应性 。

4.5. 消融 研究 和 参数 敏感性 分析

为进一步研究HerbToxNet的贡献因素，我们引入了五个变体：(i) w/o HAN直接使用草药的属性和功效数据作为特征，而不构建异构图和元路径，但结合对比学习和加权标签融合策略进行预测。(ii) w/o PE随机初始化异构图中的草药节点特征，而不是使用药理属性和功效数据作为初始特征。(iii) w/o CL移除了训练阶段具有动态系数的对比学习模块。(iv) w/o δ 用毒性标签相似度β_{ij}替换动态系数δ_{ij} 。(v) w/o WLF仅使用MLP进行预测，而不结合加权标签融合策略。详细信息报告在补充文件的第二节。总体而言，其变体明显优于其他变体，突显了每个移除因素在HerbToxNet中对草药毒性预测的重要作用。

我们还进行了不同的实验来研究 HerbToxNet 的参数敏感性。结果在补充文件的第 III 部分报告和分析。

4.6. 毒性 预测 *of* 草药

在这个案例研究中，我们利用HerbToxNet来预测和分析规范草药的毒性，包括柴胡、何首乌、巴豆、附子和白马齿苋，在现实场景中。为了严格评估模型的泛化能力，数据集中所有其他草药被用作训练集，而选定的五种草药被作为单独的测试集进行毒性预测。预测的毒性被与已知的毒性标签进行比较，并从PubMed检索相关文献，以进一步验证模型预测与当前研究成果和传统中医理论的相符性。

结果表明，HerbToxNet能够准确识别经典肝毒性草药（如柴胡和何首乌）的关键毒性，有效揭示“寒”性或“热”性草药（如巴豆和附子）的毒性风险，并揭示传统上被归类为“平”性草药（如白马齿苋）的潜在毒性。此外，HerbToxNet生成的毒性预测与既定的毒理学证据和经典的中医药分类系统高度一致，表明该模型能够利用传统草药特性（性质、味道、经度趋向）中编码的信息来评估毒性风险。

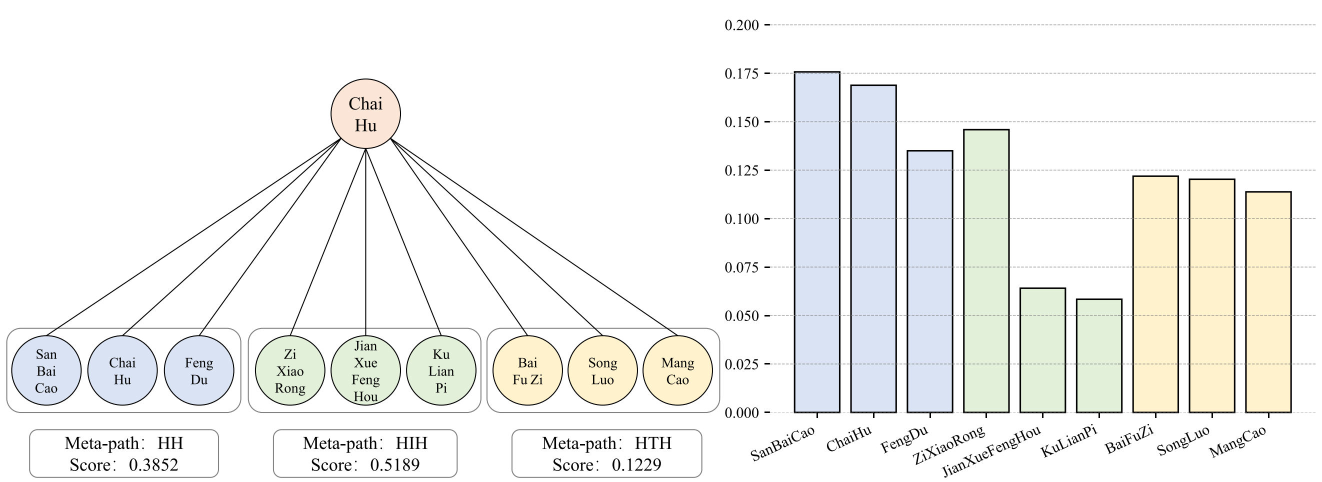
详细描述了实验程序、案例预测结果和综合分析，均在补充材料IV中提供。

4.7. 可解释性 分析

4.7.1. 模型 可解释性 分析

为了捕捉草药、成分和靶点之间的复杂关系，我们采用了异构注意力网络（HAN）（王等人，2019年），其特别之处在于其层次化注意力机制。该机制使模型能够在学习节点嵌入时捕捉节点的邻居的重要性以及不同元路径的相关性。为了更深入地分析模型的可解释性，并更好地理解邻居和元路径的相对重要性，我们对层次化注意力机制进行了详细分析。具体来说，我们以柴胡为例，并可视化不同元路径下节点与其邻居之间的注意力分数，如图2所示。从语义级层次化注意力的角度来看，HAN对三个元路径（HIH、HH和HTH）分配了不同的注意力权重，反映了模型在表示草药节点柴胡时对不同语义关系的侧重。其中，HIH元路径获得了最高的语义注意力分数，表明具有相同成分的草药在柴胡的特征学习中起主导作用。相比之下，HH元路径获得中等权重，而HTH元路径得分最低。这种注意力分数的分布与我们的数据集中的连接模式一致。HIH元路径具有较少但更具体的关联，使模型能捕捉独特的成分级特征。另一方面，HTH元路径涉及更多连接，导致更密集但更冗余的交互，降低其精确草药表示的效用。HH的中间权重反映了这些草药-草药关联是从药理和功效相似性间接推断的，因此不如HIH具体，但也不如HTH冗余。

在节点级注意力机制中，我们选择在每个元路径下注意力分数最高的三个节点。对于HH元路径，柴胡连接到三白草、自身和风毒。这三种草药都具有辛辣味的特征。在经度趋向方面，柴胡和风毒都靶向肝经。虽然三白草主要靶向肺经，但肺和肝在生理上是相互关联的，允许产生互补的治疗效果。功能上，这三种草药都表现出抗炎、抗肿和镇痛的特性。此外，与自节点的连接



(a) Semantic-level attention across meta-paths　　(b) Node-level attention under different meta-paths

图。，

also highlight the methodological adaptability of its core design across different toxicity prediction scenarios.

4.5. Ablation study and parameter sensitivity analysis

To further study the contribution factors of HerbToxNet, we introduce five variants: (i) w/o HAN directly uses the properties and efficacy data of herb as features without constructing heterogeneous graph and the meta-path, but combined with contrastive learning and weighted label fusion strategies for prediction. (ii) w/o PE randomly initializes the herb node features in the heterogeneous graph instead of using pharmacological property and efficacy data as initial features. (iii) w/o CL removes the contrastive learning module with dynamic coefficients in the training phase. (iv) w/o δ replaces the dynamic coefficient δ_{ij} with the toxicity label similarity β_{ij}. (v) w/o WLF only uses the MLP for prediction, without incorporating the weighted label fusion strategy. The details are reported in Section II of Supplementary file. Overall, outperforms its variants by a clear margin, underscoring the vital role of each removed factor in HerbToxNet for the prediction of herb toxicity.

We also conduct different experiments to study the parameter sensitivity of HerbToxNet. The results are reported and analyzed in Section III of Supplementary file.

4.6. Toxicity prediction of herbs

In this case study, we utilized HerbToxNet to predict and analyze the toxicity of canonical herbs, including *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, and *Bi Ma Zi*, in real-world scenarios. To rigorously evaluate the model's generalization capability, all other herbs in the dataset were used as the training set, while the five selected herbs were treated as a separate test set for toxicity prediction. The predicted toxicities were compared with known toxic labels, and relevant literature was retrieved from PubMed to further verify the consistency of model predictions with current research findings and traditional TCM theory.

The results demonstrate that HerbToxNet can accurately identify key toxicities of canonical hepatotoxic herbs (such as *Chai Hu* and *He Shou Wu*), effectively reveal toxicity risks in extremely "cold" or "hot" herbs (such as *Ba Dou* and *Fu Zi*), and uncover potential toxicity in herbs traditionally classified as neutral ("ping") (such as *Bi Ma Zi*). Moreover, the toxicity predictions generated by HerbToxNet are broadly consistent with both established toxicological evidence and the classical TCM classification system, indicating that the model can leverage information encoded in traditional herb properties (nature, flavor, meridian tropism) to assess toxicity risk.

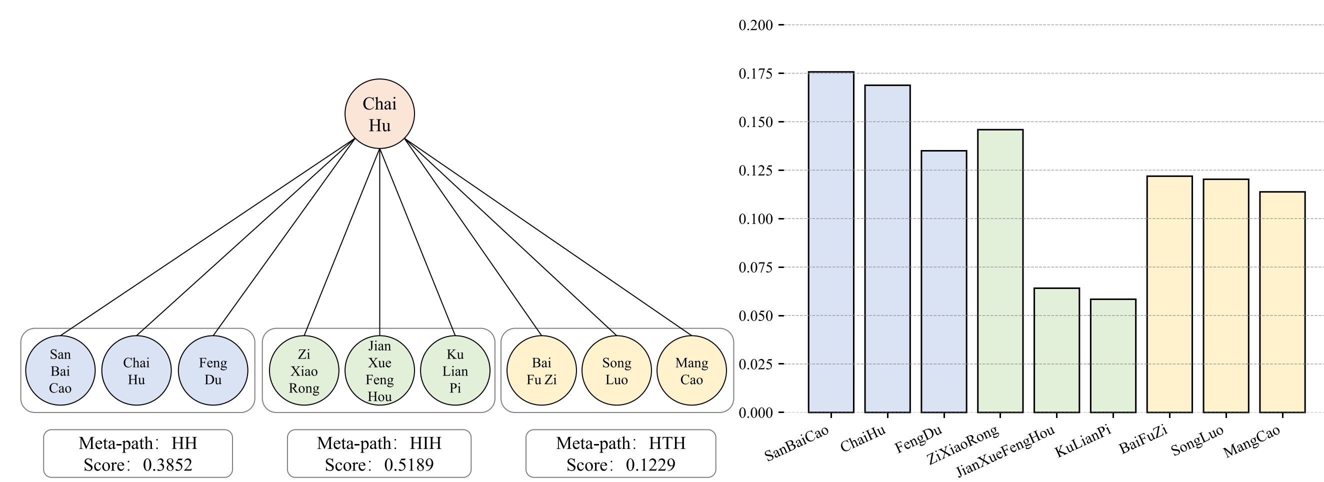
The detailed description of the experimental procedures, case prediction results, and comprehensive analyses are provided in Supplementary Section IV.

4.7. Interpretability analysis

4.7.1. Model interpretability analysis

To capture the intricate relationships between herbs, ingredients and targets, we adopt the Heterogeneous Attention Network (HAN) (Wang et al., 2019), which is particularly notable for its hierarchical attention mechanism. This mechanism enables the model to capture both the importance of a node's neighbors and the relevance of different meta-paths when learning node embeddings. To gain deeper insight into the model interpretability and to better understand the relative significance of neighbors and meta-paths, we conduct a detailed analysis of the hierarchical attention mechanism. Specifically, we take *Chai Hu* as a case study and visualize the attention scores between the node and its neighbors under different meta-paths, as illustrated in Fig. 2. From the perspective of semantic-level hierarchical attention, HAN assigns distinct attention weights to three meta-paths (HIH, HH and HTH), reflecting the model's emphasis on different semantic relationships when representing the herb node *Chai Hu*. Among them, the HIH meta-path receives the highest semantic attention score, indicating that herbs with shared ingredients play a dominant role in the feature learning of *Chai Hu*. In contrast, the HH meta-path receives a moderate weight, while the HTH meta-path has the lowest score. This distribution of attention scores aligns with the connectivity patterns in our dataset. The HIH meta-path, with fewer but more specific associations, enables the model to capture distinctive ingredient-level features. On the other hand, the HTH meta-path, involving far more connections, leads to denser but more redundant interactions, diminishing its utility for precise herb representation. The intermediate weight for HH reflects that these herb-herb associations are indirectly inferred from pharmacological and efficacy similarities, making them less specific than HIH but less redundant than HTH.

In the node-level attention mechanism, we select the top three nodes with the highest attention scores under each meta-path. For HH meta-path, *Chai Hu* is connected to *San Bai Cao*, itself, and *Feng Du*. All three herbs share the characteristic of a pungent flavor. Regarding meridian tropism, both *Chai Hu* and *Feng Du* target the liver meridian. Although *San Bai Cao* primarily targets the lung meridian, the lung and liver are physiologically interrelated, allowing for complementary therapeutic effects. Functionally, all three exhibit anti-inflammatory, anti-swelling, and analgesic properties. In addition, the connection to the self-node



(a) Semantic-level attention across meta-paths　　(b) Node-level attention under different meta-paths

Fig. 2. Hierarchical attention analysis results of HerbToxNet on the herb node *Chai Hu*, including (a) semantic-level attention across meta-paths, and (b) node-level attention under different meta-paths.

进一步 表明 该 节点 本身 在 学习 过程 中 在 生成 其 自 身 表 示 发 挥 着<style id='37'> 至关<style id='39'> 重<style id='41'> 作<style id='43'> 用<style id='45'> 。

对于HIH元路径，柴胡与紫霄茸、建雪风后和苦连皮相连。柴胡、紫霄茸、苦连皮和建雪风后共享次生代谢物的共同类别，包括三萜皂苷、黄酮类、酚类化合物、挥发油和苦生物碱。例如，柴胡含有皂苷元，紫霄茸富含黄酮类和酚酸，苦连皮含有山豆根素和相关柠檬内酯，而建雪风后则含有抗阿片碱等强心苷类生物碱。尽管它们结构多样，但这些成分具有重叠的生物活性，尤其是抗炎、抗肿瘤和解毒作用。

对于HTH元路径，柴胡与白附子、松萝和芒草相连。这些草药与柴胡共享几种潜在的药理靶点，包括TNF、IL6、PTGS2和MAPK1，这些靶点与肝损伤和炎症反应密切相关。柴胡中的皂苷元和白附子中的二萜生物碱可能通过激活NFKB信号通路上调TNF和IL6等促炎细胞因子。松萝中的乌索酸与线粒体功能障碍和氧化应激有关，导致PTGS2过表达介导的肝毒性。芒草中的生物碱可能影响MAPK1信号通路，该通路参与肝细胞凋亡和应激反应。

In 总结中， 节点级别的 注意力 机制突出了 基于共享 草药 属性、 成分、 靶点 的关键 关联， 为 模型 的 预测 提供了 具有 生物意义 和 可解释性 的 见解， 特别是针对 柴胡 胡。

4.7.2. 生物可解释性 分析

为了增强HerbToxNet的生物可解释性，我们沿着“草药-成分-靶点-生物通路-毒性”轴构建了一个多级解释框架。以柴胡为例，我们通过整合元路径分析、文献综述和毒性数据库，系统地识别了其毒性成分、相关分子靶点以及相关的毒性类型。我们进一步通过生物通路和功能注释将这些成分联系起来。结果表明，HerbToxNet成功地揭示了与肝毒性、肾毒性和全身毒性相关的关键毒性成分及其分子靶点，并揭示了草药诱导毒性反应背后的相关生物通路和机制。这表明HerbToxNet能够为其毒性预测提供机制上有意义且生物上合理的解释，从而支持草药的更可靠风险评估。分析过程的详细描述、深入结果以及所有支持证据可在补充部分V中找到。

5. 结论

在本文中，我们构建了一个多标签草药毒性数据集，并提出了HerbToxNet，这是一个集成了异构网络建模、对比学习和加权标签融合的草药毒性预测框架。实证研究表明，HerbToxNet比竞争方法具有更好的性能。HerbToxNet可以识别新的潜在毒性，并提高草药的安全临床使用。此外，它还可以揭示与毒性相关的成分和靶点，提供对毒性机制的见解，指导解毒剂开发，支持毒性相关数据库的构建和更安全的草药的发现。尽管HerbToxNet比比较方法取得了更好的性能并给出了一些真实的结果，但它仍然存在局限性。我们的数据集目前涵盖了五种毒性类别，许多草药被分配了多个标签，支持多标签分类的使用。由于标准化数据的限制，更广泛的毒性类别在本阶段未包含在内。在未来工作中，我们将扩展毒性覆盖范围，以提高HerbToxNet的鲁棒性和适用性。

CRediT 作者署名 贡献 声明

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数据 可用性

我们的 代码 和 收集的 数据集 是 共享的 在 https://www.sdu-idea.cn/codes.php?name=HerbToxNet.

利益冲突声明 的 声明

余国先报告说，财务支持由中华人民共和国科学技术部提供。如果还有其他作者，他们声明他们没有已知的利益冲突或个人关系可能影响本论文中报告的工作。

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补充 材料

补充 材料 相关 与 本 文 相关 的 补充 材料 可 在 在线 版本 中找到，

参考文献

凯撒， L. & Cech N. (2019). Synergy and antagonism in natural product extracts: when , , , , 1 + 1 does not equal 2. *Natural Product Reports 36*(6) 869–888. https://doi.org/10. /CNP A. 陈, 1039 9 00011

K. (2003). Some aspects of toxic contaminants in herbal medicines. *Chemosphere* , , 陈, T., *52*(9) 1361–1371. https://doi.org/10.1016/S0045-6535(03)00471-5.

Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for con- trastive learning of visual representations. In *Int. conf. mach. learn.* (pp. 1597–1607). 切尔卡索夫, https://doi.org/10.5555/3524938.3525087.

A. Muratov E. Fourches D. Varnek A. Baskin I. Cronin M. Dearden , , , , , , , , , , , J. Gramatica P. Martin Y. Todeschini

, , , , , , , , , //doi.org/10.1093/nar/gkaa1063. 弗洛拉 ,

S. Shrivastava R. & Mittal M. (2013). Chemistry and pharmacological properties , , , , , , 高, J., of some natural and synthetic antioxidants for heavy metal toxicity. *Current Medicinal Chemistry 20*(36) 4540–4574. https://doi.org/10.2174/09298673113209990146.

Wu, L., Lin, G., Zou, J., Yan, B., Liu, K., He, S., & Bo, X. (2025). Multi-task multi- view and iterative error-correcting random forest for acute toxicity prediction. *Expert* , , 高, *Systems with Applications 274* 126972. https://doi.org/10.1016/j.eswa.2025.126972.

Y. Liang A. Fan X. Hu L. Hao F. & Li Y. (2019). Safety research in traditional , , , , , , , , , , , chinese medicine: methods applications and outlook. *Engineering 5*(1) 76–82. https: , , , //doi.org/10.1016/j.eng.2018.11.019. 葛 ,

Z. Jiang X. Tong Z. Feng P. Zhou B. Xu M. Wang Z. & Pang Y. (2021). , , , , , , , , , , , Multi-label correlation guided feature fusion network for abnormal ECG diagno- , , , , sis. *Knowledge-Based System 233* 107508. https://doi.org/10.1016/j.knosys.2021. , , , . 107508

龚， L. Jiang J. Chen S. & Qi M. (2023). A syndrome differentiation model of TCM , , , , , , , , based on multi-label deep forest using biomedical text mining. *Frontiers in Genetics 14* , 哈德塞尔 , R., 1272016. https://doi.org/10.3389/fgene.2023.1272016.

Chopra, S., & LeCun, Y. (2006). Dimensionality reduction by learning an in- variant mapping. In *IEEE conf. comput. vis. pattern recognit. (cvpr)* (pp. 1735–1742). https://doi.org/10.1109/CVPR.2006.100.

further indicates that the node itself plays a crucial role in generating its own representation during the learning process.

For HIH meta-path, *Chai Hu* is connected to *Zi Xiao Rong*, *Jian Xue Feng Hou*, and *Ku Lian Pi*. *Chai Hu*, *Zi Xiao Rong*, *Ku Lian Pi*, and *Jian Xue Feng Hou* share common classes of secondary metabolites, which include triterpenoid saponins, flavonoids, phenolic compounds, volatile oils, and bitter alkaloids. For instance, *Chai Hu* contains saikosaponins, *Zi Xiao Rong* is rich in flavonoids and phenolic acids, *Ku Lian Pi* contains toosendanin and related limonoids, while *Jian Xue Feng Hou* harbors cardiotoxic cardiac glycosides such as antiarin. Despite their structural diversity, these ingredients share overlapped biological activities, notably anti-inflammatory, antitumor, and detoxifying effects.

For the HTH meta-path, *Chai Hu* is connected to *Bai Fu Zi*, *Song Luo*, and *Mang Cao*. These herbs share several potential pharmacological targets with *Chai Hu*, including *TNF*, *IL6*, *PTGS2*, and *MAPK1*, which are closely associated with liver injury and inflammatory responses. Saikosaponins in *Chai Hu* and diterpenoid alkaloids in *Bai Fu Zi* may upregulate pro-inflammatory cytokines such as *TNF* and *IL6* via activation of the *NFKB* signaling pathway. Usnic acid in *Song Luo* has been implicated in mitochondrial dysfunction and oxidative stress, leading to hepatotoxicity mediated by *PTGS2* overexpression. Alkaloids in *Mang Cao* may affect *MAPK1* signaling, which is involved in hepatocyte apoptosis and stress response.

In summary, the node-level attention mechanism highlights key associations based on shared herb properties, ingredients, and targets, offering biologically meaningful and interpretable insights into the model's predictions for *Chai Hu*.

4.7.2. Biointerpretability analysis

To enhance the biological interpretability of HerbToxNet, we constructed a multi-level explanatory framework along the "herb-ingredient-target-biological pathway-toxicity" axis. Taking *Chai Hu* as a representative case, we systematically identified its toxic ingredients, related molecular targets, and associated toxicity types by integrating meta-path analysis, literature curation, and toxicity databases. We further linked these components via their biological pathways and functional annotations. Our results show that HerbToxNet successfully uncovers key toxic ingredients and their molecular targets involved in hepatotoxicity, nephrotoxicity, and systemic toxicity, and reveals the relevant biological pathways and mechanisms underlying herb-induced toxic responses. This demonstrates that HerbToxNet can provide mechanistically meaningful and biologically plausible explanations for its toxicity predictions, thereby supporting more reliable risk assessment of herbs. A comprehensive description of the analytical procedures, in-depth results, and all supporting evidence can be found in Supplementary Section V.

5. Conclusion

In this paper, we construct a multi-label herb toxicity dataset and propose HerbToxNet, a framework for herb toxicity prediction that integrates heterogeneous network modeling, contrastive learning, and weighted label fusion. Empirical study shows that HerbToxNet achieves superior performance than competitive methods. HerbToxNet can identify novel potential toxicities and improve the safe clinical use of herbs. Moreover, it can uncover toxicity-related ingredients and targets, provide insights into toxicity mechanisms, guide antidote development, support toxicity-related databases construction and the discovery of safer herbs . Although HerbToxNet achieves better performance than compared methods and gives some authentic results, it still has limitations. Our dataset currently covers five toxicity categories, with many herbs assigned multiple labels, supporting the use of multi-label classification. Due to limited standardized data, broader toxicity categories are not included at this stage. In future work, we will expand toxicity coverage to enhance the robustness and applicability of HerbToxNet.

CRediT authorship contribution statement

Yongzheng Zhu: Data curation, Formal analysis, Investigation, Software, Validation, Writing – original draft; **Yunbo Miao:** Formal analysis, Software, Validation, Writing – original draft; **Rong Sun:** Supervision, Resources; **Zhongmin Yan:** Methodology, Formal analysis, Writing – review & editing; **Guoxian Yu:** Supervision, Investigation, Methodology, Formal analysis, Writing – review & editing.

Data Availability

Our code and collected dataset are shared at https://www.sdu-idea.cn/codes.php?name=HerbToxNet.

Declaration of competing interest

Guoxian Yu reports financial support was provided by Ministry of Science and Technology of the People's Republic of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.eswa.2025.129969.

References

Caesar, L., & Cech, N. (2019). Synergy and antagonism in natural product extracts: when 1 + 1 does not equal 2. *Natural Product Reports*, *36*(6), 869–888. https://doi.org/10.1039/C9NP00011A.

Chan, K. (2003). Some aspects of toxic contaminants in herbal medicines. *Chemosphere*, *52*(9), 1361–1371. https://doi.org/10.1016/S0045-6535(03)00471-5.

Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *Int. conf. mach. learn.* (pp. 1597–1607). https://doi.org/10.5555/3524938.3525087.

Cherkasov, A., Muratov, E., Fourches, D., Varnek, A., Baskin, I., Cronin, M., Dearden, J., Gramatica, P., Martin, Y., Todeschini, R. et al. (2014). Qsar modeling: Where have you been? where are you going to? *Journal of Medicinal Chemistry*, *57*(12), 4977–5010. https://doi.org/10.1021/jm4004285.

Fang, S., Dong, L., Liu, L., Guo, J., Zhao, L., Zhang, J., Bu, D., Liu, X., Huo, P., Cao, W. et al. (2021). Herb: A high-throughput experiment-and reference-guided database of traditional chinese medicine. *Nucleic Acids Research*, *49*(D1), D1197–D1206. https://doi.org/10.1093/nar/gkaa1063.

Flora, S., Shrivastava, R., & Mittal, M. (2013). Chemistry and pharmacological properties of some natural and synthetic antioxidants for heavy metal toxicity. *Current Medicinal Chemistry*, *20*(36), 4540–4574. https://doi.org/10.2174/09298673113209990146.

Gao, J., Wu, L., Lin, G., Zou, J., Yan, B., Liu, K., He, S., & Bo, X. (2025). Multi-task multi-view and iterative error-correcting random forest for acute toxicity prediction. *Expert Systems with Applications*, *274*, 126972. https://doi.org/10.1016/j.eswa.2025.126972.

Gao, Y., Liang, A., Fan, X., Hu, L., Hao, F., & Li, Y. (2019). Safety research in traditional chinese medicine: methods, applications, and outlook. *Engineering*, *5*(1), 76–82. https://doi.org/10.1016/j.eng.2018.11.019.

Ge, Z., Jiang, X., Tong, Z., Feng, P., Zhou, B., Xu, M., Wang, Z., & Pang, Y. (2021). Multi-label correlation guided feature fusion network for abnormal ECG diagnosis. *Knowledge-Based System*, *233*, 107508. https://doi.org/10.1016/j.knosys.2021.107508.

Gong, L., Jiang, J., Chen, S., & Qi, M. (2023). A syndrome differentiation model of TCM based on multi-label deep forest using biomedical text mining. *Frontiers in Genetics*, *14*, 1272016. https://doi.org/10.3389/fgene.2023.1272016.

Hadsell, R., Chopra, S., & LeCun, Y. (2006). Dimensionality reduction by learning an invariant mapping. In *IEEE conf. comput. vis. pattern recognit. (cvpr)* (pp. 1735–1742). https://doi.org/10.1109/CVPR.2006.100.

He, S., Zhang, C., Zhou, P., Zhang, X., Ye, T., Wang, R., Sun, G., & Sun, X. (2019a). Herb-induced liver injury: Phylogenetic relationship, structure-toxicity relationship, and herb-ingredient network analysis. *International Journal of Molecular Sciences*, *20*(15), 3633. https://doi.org/10.3390/ijms20153633.

He, S., Zhang, X., Lu, S., Zhu, T., Sun, G., & Sun, X. (2019b). A computational toxicology approach to screen the hepatotoxic ingredients in traditional chinese medicines: Polygonum multiflorum thunb as a case study. *Biomolecules*, *9*(10), 577. https://doi.org/10.3390/biom9100577.

Hu, H., Li, Y., Zheng, Z., Hu, W., Lin, R., & Kang, Y. (2025). A traditional chinese medicine prescription recommendation model based on contrastive pre-training and hierarchical structure network. *Expert Systems with Applications*, *268*, 126318. https://doi.org/10.1016/j.eswa.2024.126318.

Jia, C., Li, X., Hu, S., Liu, G., Fang, J., Zhou, X., Yan, X., & Yan, B. (2024). Advanced mass-spectra-based machine learning for predicting the toxicity of traditional chinese medicines. *Analytical Chemistry*, *97*(1), 783–792. https://doi.org/10.1021/acs.analchem.4c05311.

Jiang, M., Yang, J., Zhang, C., Liu, B., Chan, K., Cao, H., & Lu, A. (2010). Clinical studies with traditional chinese medicine in the past decade and future research and development. *Planta Medica*, *76*(17), 2048–2064. https://doi.org/10.1055/s-0030-1250456.

Jităreanu, A., Trifan, A., Vieriu, M., Caba, I.-C., Mârțu, I., & Agoroaei, L. (2022). Current trends in toxicity assessment of herbal medicines: A narrative review. *Processes*, *11*(1), 83. https://doi.org/10.3390/pr11010083.

Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., & Krishnan, D. (2020). Supervised contrastive learning. In *Adv. neural inf. process. syst.* (pp. 18661–18673). https://doi.org/10.5555/3495724.3497291.

Li, T., Tong, W., Roberts, R., Liu, Z., & Thakkar, S. (2020). DeepDILI: Deep learning-powered drug-induced liver injury prediction using model-level representation. *Chemical Research in Toxicology*, *34*(2), 550–565. https://doi.org/10.1021/acs.chemrestox.0c00374.

Li, Y., Zhang, Y., Wang, Y., Li, Y., Yang, F., Zhang, P., Zhang, Y., & Liu, C. (2019). A strategy for the discovery and validation of toxicity quality marker of chinese medicine based on network toxicology. *Phytomedicine*, *54*, 365–370. https://doi.org/10.1016/j.phymed.2018.01.018.

Liu, R., Li, X., Huang, N., Fan, M., & Sun, R. (2020). Toxicity of traditional Chinese medicine herbal and mineral products. *Advances in Pharmacology*, *87*, 301–346. https://doi.org/10.1016/bs.apha.2019.08.001.

Lv, Q., Chen, G., He, H., Yang, Z., Zhao, L., Zhang, K., & Chen, C. Y. C. (2023). Tcmbank-the largest tcm database provides deep learning-based chinese-western medicine exclusion prediction. *Signal Transduction and Targeted Therapy*, *8*(1), 127. https://doi.org/10.1038/s41392-023-01339-1.

Pham, T., Tao, X., Zhang, J., Yong, J., Li, Y., & Xie, H. (2022). Graph-based multi-label disease prediction model learning from medical data and domain knowledge. *Knowledge-Based System*, *235*, 107662. https://doi.org/10.1016/j.knosys.2021.107662.

Qiao, Y., Zhang, Y., Peng, S., Huo, M., Li, J., Cao, Y., & Chen, Z. (2022). Property theory of chinese materia medica: Clinical pharmacodynamics of traditional chinese medicine. *Journal of Traditional Chinese Medical Sciences*, *9*(1), 7–12. https://doi.org/10.1016/j.jtcms.2022.01.006.

Qiu, S., Wang, M., Yang, Y., Yu, G., Wang, J., Yan, Z., Domeniconi, C., & Guo, M. (2023). Meta multi-instance multi-label learning by heterogeneous network fusion. *An International Journal on Information Fusion*, *94*, 272–283. https://doi.org/10.1016/j.inffus.2023.02.010.

Raies, A., & Bajic, V. (2016). In silico toxicology: Computational methods for the prediction of chemical toxicity. *Wiley Interdisciplinary Reviews Computational Molecular Science*, *6*(2), 147–172. https://doi.org/10.1002/wcms.1240.

Ridnik, T., Ben-Baruch, E., Zamir, N., Noy, A., Friedman, I., Protter, M., & Zelnik-Manor, L. (2021). Asymmetric loss for multi-label classification. In *IEEE/CVF int. conf. comput. vis.* (pp. 82–91). https://doi.org/10.1109/ICCV48922.2021.00015.

Silberg, J., Swanson, K., Simon, E., Zhang, A., Ghazizadeh, Z., Ogden, S., Hamadeh, H., & Zou, J. (2024). Unitox: Leveraging LLMs to curate a unified dataset of drug-induced toxicity from FDA labels. *Advances in Neural Information Processing Systems*, *37*, 12078–12093. https://doi.org/10.1101/2024.06.21.24309315.

Sun, Y., Shi, S., Li, Y., & Wang, Q. (2019). Development of quantitative structure-activity relationship models to predict potential nephrotoxic ingredients in traditional chinese medicines. *Food and Chemical Toxicology*, *128*, 163–170. https://doi.org/10.1016/j.fct.2019.03.056.

Wang, X., Ji, H., Shi, C., Wang, B., Ye, Y., Cui, P., & Yu, P. (2019). Heterogeneous graph attention network. In *Www conf.* (pp. 2022–2032). https://doi.org/10.1145/3308558.3313562.

Wang, X., Qu, H., Liu, P., & Cheng, Y. (2004). A self-learning expert system for diagnosis in traditional chinese medicine. *Expert Systems with Applications*, *26*(4), 557–566. https://doi.org/10.1016/j.eswa.2003.10.004.

Wang, Z., Li, L., Yan, J., & Yao, Y. (2020). Approaching high-accuracy side effect prediction of traditional chinese medicine compound prescription using network embedding and deep learning. *IEEE Access*, *8*, 82493–82499. https://doi.org/10.1109/ACCESS.2020.2991750.

Wei, J., Dai, J., Sun, Y., Meng, Z., Ma, H., & Zhou, Y. (2024). Tirpnet: Risk prediction of traditional chinese medicine ingredients based on a deep neural network. *Journal of Ethnopharmacology*, *325*, 117860. https://doi.org/10.1016/j.jep.2024.117860.

Wu, Q., Cai, C., Guo, P., Chen, M., Wu, X., Zhou, J., Luo, Y., Zou, Y., Liu, A., Wang, Q. et al. (2019a). In silico identification and mechanism exploration of hepatotoxic ingredients in traditional chinese medicine. *Frontiers in Pharmacology*, *10*, 458. https://doi.org/10.3389/fphar.2019.00458.

Wu, Q., Yin, C., Li, Y., Cai, J., Yang, H., Huang, Y., Zheng, Y., Xiong, K., Yu, H., Lu, A. et al. (2021). Detecting critical functional ingredients group and mechanism of xuebijing injection in treating sepsis. *Frontiers in Pharmacology*, *12*, 769190. https://doi.org/10.3389/fphar.2021.769190.

Wu, Y., Zhang, F., Yang, K., Fang, S., Bu, D., Li, H., Sun, L., Hu, H., Gao, K., Wang, W. et al. (2019b). SymMap: An integrative database of traditional chinese medicine enhanced by symptom mapping. *Nucleic Acids Research*, *47*(D1), D1110–D1117. https://doi.org/10.1093/nar/gky1021.

Xu, X., Xu, H., Shang, Y., Zhu, R., Hong, X., Song, Z., & Yang, Z. (2021). Development of the general chapters of the chinese pharmacopoeia 2020 edition: A review. *Journal of Pharmaceutical Analysis*, *11*(4), 398–404. https://doi.org/10.1016/j.jpha.2021.05.001.

Yan, B., Ye, X., Wang, J., Han, J., Wu, L., He, S., Liu, K., & Bo, X. (2022a). An algorithm framework for drug-induced liver injury prediction based on genetic algorithm and ensemble learning. *Molecules*, *27*(10), 3112. https://doi.org/10.3390/molecules27103112.

Yan, D., Zheng, G., Wang, C., Chen, Z., Mao, T., Gao, J., Yan, Y., Chen, X., Ji, X., Yu, J. et al. (2022b). Hit 2.0: An enhanced platform for herbal ingredients' targets. *Nucleic Acids Research*, *50*(D1), D1238–D1243. https://doi.org/10.1093/nar/gkab1011.

Yang, J., Wang, S., Zhang, T., Sun, Y., Han, L., Banahene, P. O., & Wang, Q. (2021). Predicting the potential toxicity of 26 components in cassiae semen using in silico and in vitro approaches. *Current Research in Toxicology*, *2*, 237–245. https://doi.org/10.1016/j.crtox.2021.06.001.

Yang, K., Zhou, X., Zhang, R., Liu, B., Lei, L., Zhang, X., Chu, H., Sun, C., Gao, Z., & Xu, H. (2015). Integrating herb effect similarity for network-based herb target prediction. In *Int. conf. biomed. eng. inform.* (pp. 483–488). https://doi.org/10.1109/BMEI.2015.7401553.

Yu, Z., Kou, F., Gao, Y., Gao, F., Lyu, C.-m., & Wei, H. (2025). A machine learning model for predicting abnormal liver function induced by a Chinese herbal medicine preparation (Zhengqing fengtongning) in patients with rheumatoid arthritis based on real-world study. *Journal of Integrative Medicine*, *23*(1), 25–35. https://doi.org/10.1016/j.joim.2024.12.001.

Zhu, Y., Ren, L., Sun, R., Wang, J., & Yu, G. (2025). Herb-target interaction prediction by multi-instance learning. *IEEE Transactions on Artificial Intelligence*, *6*(5), 1184–1193. https://doi.org/10.1109/TAI.2024.3515932.